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The Liouville–Arnold theorem for symplectic mechanical systems

MASTER’S THESIS

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and Mathematics**



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Abstract

The Lagrangian and Hamiltonian formalisms from classical mechanics admit a formulation in the language of differential geometry, where symplectic geometry is especially well suited to encode Hamilton's equations in a coordinate-free way. Within this geometric framework, the Liouville–Arnold theorem gives sufficient conditions for the equations of motion of a Hamiltonian system to be completely integrable. In this master's thesis, a self-contained proof of this important result is presented. To this end, we start reviewing the foundations of symplectic geometry and its applications to Hamiltonian mechanics. After that, we introduce the notion of integrable system and state and prove the Liouville–Arnold theorem. Finally, various examples are used to illustrate the theory, such as Liouville tori and action-angle variables.

Keywords:

completely integrable Hamiltonian systems, Liouville–Arnold theorem, action-angle variables, symplectic manifolds, Hamiltonian vector fields, Poisson brackets

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1. Introduction

1.1 Geometric mechanics

Analytical mechanics begins with Newton's *Principia* (1687), where he first used the tools of calculus to state and develop his three famous laws of motion. Among other achievements, Newton managed to explain Kepler's laws of planetary motion, which Kepler had obtained empirically. The study of analytical mechanics experienced a substantial advance during the 18th and 19th centuries thanks to the contribution of many mathematicians, such as Euler, Hamilton, Jacobi, Lagrange, Legendre, and Poisson, among others. Their techniques are essentially based on the application of variational methods to obtain the equations of motion of dynamical systems. For example, Hamilton's principle leads to the Euler–Lagrange equations and the Lagrangian formalism of mechanics.

In the second half of the 20th century, an extensive group of mathematicians and physicists used the techniques of differential geometry to formalize analytical mechanics and study the properties of the dynamical systems involved intrinsically, giving rise to what is known today as geometric mechanics. In earlier times, the only techniques available were analytical. By obtaining a complete or approximate quantitative solution, qualitative or geometric properties could be deduced. It was Poincaré's idea to obtain qualitative information using qualitative methods, that is, geometric methods. This new approach was then developed by the works of S. Smale, G. Reeb, V. I. Arnold, J. Moser, J. Marsden, and many others. The use of tools from differential geometry, such as the tangent and cotangent bundles, vector fields, differential forms, tensor fields, Riemannian metrics, and Lie groups, revolutionized the field of mechanics. These topics are widely covered in the literature; see [1, 11, 12].

In particular, a new branch of differential geometry emerged, where Hamilton's equations for an autonomous system are most naturally expressed. It was given the name of *symplectic geometry* (see [1, 3, 4, 11] for more details). In the symplectic framework, the phase space of a Hamiltonian system is naturally identified with the cotangent bundle of the configuration space, and Hamilton's equations are compactly expressed in terms of the so-called symplectic form.

1.2 Integrability

The concept of integrability in classical mechanics refers to the remarkable property of certain mechanical systems whose equations of motion can be solved completely. When Newton formulated his laws of motion, the first celebrated example of an integrable system emerged naturally: the two-body problem, in which two celestial objects subject to gravity orbit around each other. This system can be solved exactly, yielding Kepler's elliptical orbits as a consequence. For many years, mathematicians and physicists tried to integrate the three-body problem, adding one planet to the previous problem. However, Poincaré ended up proving that the three-body problem is generically non-integrable, which reflects the non-triviality of the concept of integrability. Other examples of integrable systems are the harmonic oscillator, the simple pendulum, and some simple rigid body systems such as the Euler top. However, when we add complexity to a system, it usually loses this property. Apart from the three-body problem, an instructive example is the double pendulum. See [1] for more details.

In 1853, J. Liouville proved one of the most important results on integrability. In a Hamiltonian system with n degrees of freedom, if n independent conserved quantities are known, the equations of motion can be integrated. Liouville stated this theorem in its local version, that is, considering $\mathbb{R}^{2n}$ as

the phase space. In 1974, V. I. Arnold published *Mathematical Methods of Classical Mechanics* [2], where, using tools from differential geometry, he generalized this result to the setting of symplectic manifolds. Since then, it has been known as the Liouville–Arnold theorem, and it is considered one of the most important theorems of mechanics regarding integrability. The Liouville–Arnold theorem is covered in most modern books on mechanics; see [1, 3, 11, 13].

The practical importance of integrability is considerable: integrable systems serve as the backbone of perturbation theory, since real-world systems are often small deformations of integrable ones, and the Kolmogorov–Arnold–Moser theorem precisely quantifies how much of the integrable structure survives such perturbations. Integrability is also present in more recent research lines, such as generalizations to contact geometry [8, 9] and bi-Hamiltonian systems [6], or algebraic approaches involving Nijenhuis and Haantjes endomorphisms [14].

1.3 Objectives

The first objective of this work is to provide a self-contained introduction to symplectic geometry, requiring only a basic background in differential geometry. To this end, we thoroughly review the concepts that will set the ground for the Liouville–Arnold theorem, such as Hamiltonian vector fields and Poisson brackets.

The second and main objective of this dissertation is to give a complete and detailed explanation of the Liouville–Arnold theorem. Special care is taken in providing an elementary and highly detailed proof of the theorem, which is divided into many parts that need to be addressed with different techniques. Various lemmas and intermediate results that are often omitted in the literature are included in an attempt to make the exposition as self-contained as possible. We also try to be very explicit throughout the proof and avoid omitting intermediate steps.

Finally, this dissertation also has an applied component. Beyond the theoretical development of symplectic geometry and integrable systems, numerical simulation and computer graphics are used to visualize the geometric structures and dynamical behavior arising in classical mechanics. These visualizations were used to illustrate abstract concepts such as phase space, Hamiltonian flows, and invariant manifolds in various examples throughout the text. The code used to generate all figures and animations included in the thesis can be consulted in the GitHub repository <https://github.com/rqdefrutos/masters-thesis/>.

1.4 Structure of the dissertation

The structure of this master’s thesis is as follows. In Chapter 2, we review the basic ideas of geometric mechanics. At the same time, it serves as a brief historical journey through the ideas of Newton, Lagrange and Hamilton. In Chapter 3, we give a solid introduction to symplectic geometry, which sets the framework for the next Chapter. This introduction includes key concepts such as Darboux coordinates, symplectomorphisms, Hamiltonian vector fields and Poisson brackets. Finally, Chapter 4 is devoted to the Liouville–Arnold theorem. After introducing integrable systems, we give a detailed exposition of this important result, including a careful proof and various illustrative examples. In Appendix A we prove a key intermediate result of Chapter 4.

The notations used throughout the thesis are mostly standard. The Einstein summation convention is used whenever it applies. We always work in the category of smooth manifolds, meaning that all functions, vector fields, forms, etc., that appear throughout the text are smooth, even when we do not specify. The term *smooth* is used in the sense of *infinitely differentiable*. As a reference for

topics of differential geometry and differential topology, we used [7, 10, 15].

The main references used for this master's thesis are the books *Foundations of Mechanics* by R. Abraham and J. Marsden [1], *Mathematical Methods of Classical Mechanics* by V. I. Arnold [2], and *Integrable Hamiltonian Systems* by A. V. Bolsinov and A. T. Fomenko [3].

2. Geometric approach to classical mechanics

In this chapter, we review how differential geometry can be used to formalize the historical approaches of Newton, Lagrange and Hamilton (among many others) to classical mechanics. This approach has the advantage of providing a coordinate-free way of writing equations, thus revealing the mathematical structure of mechanical systems. For a comprehensive exposition of geometric mechanics, we refer the reader to [12].

2.1 From Newton to Lagrange

Mechanics begins with Newton's second law, which describes how a particle behaves in presence of a force. The coordinates of the particle (or particles) are the variables of the system, which live in the configuration space. From the point of view of geometric mechanics, the configuration space Q is a smooth manifold whose dimension represents the number of degrees of freedom. Since we want Q to describe physical reality, a metric is needed, so we set Q to be a Riemannian manifold.

In this general setting, the force of the system is given by a vector field $F \in \mathfrak{X}(Q)$ and Newton's equations read

$$F \circ c = \nabla_{c'} c',$$

where ∇ refers to the Levi-Civita connection of Q , induced by the metric. This means that, along a curve c , F must coincide with the covariant derivative of c' , i.e., the acceleration of c . In local coordinates q^i , the curve $c(t) = (q^1(t), \dots, q^n(t))$ satisfies

$$F^k = \ddot{q}^k + \Gamma_{ij}^k \dot{q}^i \dot{q}^j,$$

where Γ_{ij}^k are the Christoffel symbols of the metric. The masses of the particles, apparently missing in this equation, can be encoded in the metric of Q .

Let us see how the Lagrangian and Hamiltonian formalisms arise, starting with the familiar case of a conservative force. This means that $F = -\nabla V$ for some function $V \in \mathcal{C}^\infty(Q)$ (the negative sign is a convention), where ∇ denotes the gradient with respect to the given metric. The function V is called the potential energy of the system (and is defined up to a constant).

In the setting of geometric mechanics, velocities are interpreted as tangent vectors, so the kinetic energy is defined as a function on the tangent bundle TQ . Specifically, it is defined as

$$T(q, \dot{q}) = \frac{1}{2} \|\dot{q}\|_q^2,$$

where $\|\cdot\|_q$ is the norm induced by the metric in the tangent space $T_q Q$. In terms of natural coordinates $q^i, \dot{q}^i$, the expression of T is

$$T = \frac{1}{2} g_{ij} \dot{q}^i \dot{q}^j,$$

where $g_{ij} = g_{ij}(q^1, \dots, q^n)$ are the components of the metric in these coordinates.

Now we are ready to define the Lagrangian $\mathcal{L} \in \mathcal{C}^\infty(TQ)$:

$$\mathcal{L}(q, \dot{q}) = T(q, \dot{q}) - V(q)$$

or, more briefly, as $\mathcal{L} = T - \pi^* V$ where $\pi : TQ \rightarrow Q$ is the projection and $\pi^* V = V \circ \pi$ is the pullback.

On the other hand, *mechanical energy* is the function $E = T + \pi^*V \in \mathcal{C}^\infty(TQ)$. For simplicity, we usually write $\mathcal{L} = T - V$ and $E = T + V$. As is well-known, the mechanical energy is conserved, meaning that if c is a solution of the dynamical system, then $E(c(t), c'(t))$ is constant. This can be checked using properties of the covariant derivative or by direct computation in local coordinates.

The main advantage of the Lagrangian formalism is that it allows to express the dynamical equations without explicit mention of the components of the force field nor the Christoffel symbols. Newton equations turn into Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) - \frac{\partial \mathcal{L}}{\partial q^i} = 0$$

2.2 From Lagrange to Hamilton

There exists yet another formalism which uses $2n$ first-order differential equations, instead of the n second-order Euler-Lagrange equations: the Hamiltonian formalism. Let us see how to arrive to Hamilton equations starting from the Lagrangian formalism that we introduced in the previous section. First, we define the generalized momenta with respect to the coordinates q^i as

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}.$$

The potential energy V does not depend on the $\dot{q}^i$, so

$$p_i = \frac{\partial}{\partial \dot{q}^i} \left(\frac{1}{2} g_{ab} \dot{q}^a \dot{q}^b \right) = \frac{1}{2} g_{ab} (\delta_i^a \dot{q}^b + \dot{q}^a \delta_i^b) = \frac{1}{2} (g_{ib} \dot{q}^b + g_{ai} \dot{q}^a) = g_{ij} \dot{q}^j.$$

In particular, we can solve for the $\dot{q}^i$ using the inverse matrix (g^{ij}) as

$$\dot{q}^i = g^{ij} p_j,$$

so

$$T = \frac{1}{2} g_{ab} \dot{q}^a \dot{q}^b = \frac{1}{2} g_{ab} g^{ai} p_i g^{bj} p_j = \frac{1}{2} \delta_b^i p_i g^{bj} p_j = \frac{1}{2} g^{ij} p_i p_j$$

Since T is a function of position and momenta, so is the mechanical energy E , namely $E(q, \dot{q}) = \mathcal{H}(q, p(q, \dot{q}))$ where

$$\mathcal{H}(q, p) = \frac{1}{2} g^{ij}(q) p_i p_j + V(q).$$

This is the so-called Hamiltonian function. It turns out that the dynamics in this coordinates obey Hamilton equations:

$$\begin{cases} \frac{dq^i}{dt} = \frac{\partial \mathcal{H}}{\partial p_i} \\ \frac{dp_i}{dt} = -\frac{\partial \mathcal{H}}{\partial q^i} \end{cases}$$

Being essentially the energy of the system, a key feature of the Hamiltonian is that it is conserved along the integral curves.

So far, we have introduced the momenta with respect to a system of coordinates. Let us see what happens when we change coordinates:

$$\bar{p}_j = \frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}^j} = \frac{\partial \mathcal{L}}{\partial q^i} \frac{\partial q^i}{\partial \dot{\bar{q}}^j} + \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial \dot{\bar{q}}^j} = p_i \frac{\partial q^i}{\partial \dot{\bar{q}}^j}$$

where we used that

$$\frac{\partial q^i}{\partial \dot{q}^j} = 0, \quad \frac{\partial \dot{q}^i}{\partial \dot{q}^j} = \frac{\partial q^i}{\partial \dot{q}^j}.$$

This shows that the functions p_i change as the components of a 1-form, which suggests that the function $\mathcal{H}$ can be defined as a function on the cotangent bundle T^*Q , whose natural coordinates we denote (q, p) .

2.3 Beyond conservative systems

The Lagrangian formalism also makes sense in more generality, not only for conservative systems. A Lagrangian dynamical system is any pair $(TQ, \mathcal{L})$ where Q represents the configuration space of the physical system and $\mathcal{L} \in \mathcal{C}^\infty(TQ)$ is the Lagrangian function. Working in natural coordinates $q, \dot{q}$, we can still define the generalized momenta as

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}(q, \dot{q}).$$

According to the implicit function theorem, we can solve for the $\dot{q}^i$ in terms of the p_i as long as the Jacobian matrix of $\frac{\partial \mathcal{L}}{\partial \dot{q}^i}$ with respect to $\dot{q}$ is nonsingular, i.e., as long as

$$\det \left(\frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} \right) \neq 0.$$

If this happens for any set of natural coordinates, $(TQ, \mathcal{L})$ is called a *regular* Lagrangian dynamical system. These are the most studied Lagrangian systems because regularity is a sufficient condition for the existence and uniqueness of a vector field $X_{\mathcal{L}} \in \mathfrak{X}(TQ)$ whose integral curves are given (in local coordinates) by the classical Euler-Lagrange equations.

Associated with the Lagrangian is the energy function $E_{\mathcal{L}} \in \mathcal{C}^\infty(TQ)$, whose local expression is

$$E_{\mathcal{L}} = \dot{q}^i \frac{\partial \mathcal{L}}{\partial \dot{q}^i} - \mathcal{L}$$

Using the Euler-Lagrange equations, it is seen that $E_{\mathcal{L}}$ is a conserved quantity. (In the case of a conservative system, we recover the formula $E_{\mathcal{L}} = T + V$.) Having solved for the $\dot{q}^i$, we can define the local Hamiltonian function as

$$\mathcal{H}(q, p) = p_i \dot{q}^i(q, p) - \mathcal{L}(q, \dot{q}(q, p))$$

so that $E_{\mathcal{L}}(q, \dot{q}) = \mathcal{H}(q, p(q, \dot{q}))$, as before. Once again, the trajectories of the system are described by Hamilton equations.

Actually, regularity is equivalent to the *fiber derivative* of $\mathcal{L}$,

$$\mathcal{FL} : TQ \rightarrow T^*Q$$

being a local diffeomorphism. The map $\mathcal{FL}$ is called the *Legendre transformation* of $\mathcal{L}$. Using natural coordinates $q, \dot{q}$ and q, p in the tangent and cotangent bundles, respectively,

$$\mathcal{FL}(q, \dot{q}) = \left(q, \frac{\partial \mathcal{L}}{\partial \dot{q}} \right)$$

However, for $\mathcal{H}$ to be globally defined we need a stronger condition. $\mathcal{FL}$ needs to be a global diffeomorphism. In this case, the Lagrangian is called *hyperregular* and there exists a function $\mathcal{H} \in \mathcal{C}^\infty(T^*Q)$ (called Hamiltonian function) such that

$$\mathcal{FL}^* \mathcal{H} = E_{\mathcal{L}}.$$

3. Symplectic geometry

In this chapter, we give an introduction to symplectic geometry. After establishing the linear theory, we review many concepts that are essential for Hamiltonian mechanics, such as Darboux coordinates, Hamiltonian vector fields and Poisson brackets. The chapter features numerous illustrative examples and the proof of most results is included. For a comprehensive text on symplectic geometry, see [4]. We also used the introductory chapters of [1, 3] for reference.

3.1 Symplectic vector spaces

Definition 3.1.1. Let E be a finite-dimensional real vector space. A *symplectic inner product* on E is a non-degenerate skew-symmetric bilinear form $\omega : E \times E \rightarrow \mathbb{R}$.

We say that E has been endowed with a *linear symplectic structure* and that the pair (E, ω) is a *symplectic vector space*.

Recall that non-degeneracy means that if $\omega(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{y} \in E$, then $\mathbf{x} = 0$, and skew-symmetry means that $\omega(\mathbf{x}, \mathbf{y}) = -\omega(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in E$. In particular, $\omega(\mathbf{x}, \mathbf{x}) = 0$ for all $\mathbf{x} \in E$.

Let $\{\mathbf{e}_i\}$ be a basis of E and $\{\alpha^i\}$ its dual basis. In terms of this basis, ω can be expressed as

$$\omega = \omega_{ij} \alpha^i \otimes \alpha^j$$

where $\omega_{ij} = \omega(\mathbf{e}_i, \mathbf{e}_j)$. The skew symmetry implies that the matrix $A = (\omega_{ij})$ is skew-symmetric, i.e., $A = -A^\top$. Being an alternating form, ω can also be expressed with the wedge product as

$$\omega = \sum_{i < j} \omega_{ij} \alpha^i \wedge \alpha^j,$$

using only the upper triangular half of the matrix A .

Remark 3.1.2. For a linear symplectic structure to exist it is necessary that the dimension of E is even. Indeed, with the previous notations, if $r = \dim E$, then

$$\det A = \det(-A^\top) = (-1)^r \det A.$$

Since ω is non-degenerate, $\det A \neq 0$, so r must be even. Thus, from now on, we will write $\dim E = 2n$.

With a symplectic form we can give a notion of *orthogonality*.

Definition 3.1.3. Let (E, ω) be a symplectic vector space. Given a vector subspace $F \subset E$, the *symplectic orthogonal* of F is defined

$$F^\perp = \{\mathbf{x} \in E : \omega(\mathbf{x}, \mathbf{y}) = 0, \quad \mathbf{y} \in F\}.$$

Note that, for this orthogonal, it is not true in general that $F \oplus F^\perp = E$. Indeed, consider $E = \mathbb{R}^2$ with the canonical basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ and the symplectic form

$$\omega = \alpha^1 \wedge \alpha^2.$$

If we let $F = \text{span}(\mathbf{e}_1)$, the symplectic orthogonal $F^\perp$ is given by all vectors $\mathbf{x} = (x_1, x_2)$ such that $\omega(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{y} \in F$. Using that F is spanned by $\mathbf{e}_1 = (1, 0)$, we have

$$\omega((x_1, x_2), (1, 0)) = -x_2.$$

And for this to be zero, we must have $x_2 = 0$. This implies that any vector in $F^\perp$ must be of the form $(x_1, 0)$, so $F^\perp = \text{span}(\mathbf{e}_1) = F$.

Let us introduce the main types of linear subspaces of a symplectic vector space.

Definition 3.1.4. Let (E, ω) be a symplectic vector space and let $F \subset E$ be a linear subspace.

1. F is *isotropic* if $F \subset F^\perp$, that is, $\omega|_{F \times F} = 0$.
2. F is *Lagrangian* if F is isotropic and there exists an isotropic complement, that is, there exists a subspace $F' \subset E$ such that $F \oplus F' = E$.
3. F is *symplectic* if $\omega|_{F \times F}$ is a symplectic inner product on F .

It is easy to see that F is a Lagrangian subspace if and only if $F = F^\perp$. Since $\dim F + \dim F^\perp = \dim E$, it follows that F is Lagrangian if and only if it is isotropic and $\dim F = \frac{1}{2} \dim E$.

Next, we will show that every symplectic inner product ω admits a basis $\{\mathbf{e}_i\}$ in which the matrix of ω is

$$J = \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix} \quad (3.1)$$

Such a basis is called a *symplectic basis* for ω .

Lemma 3.1.5. Let E be a real vector space with $\dim E = 2n$. Every symplectic inner product ω admits a symplectic basis $\{\mathbf{e}_i\}$. Furthermore, if $\{\alpha^i\}$ is its dual basis, then

$$\omega = \sum_{i=1}^n \alpha^i \wedge \alpha^{i+n}.$$

Proof. For the first part, we proceed by induction on n . For $n = 1$, choose a nonzero vector $\mathbf{e}_1 \in E$. Since ω is non-degenerate, there exists a vector $\mathbf{e}_2$ such that $\omega(\mathbf{e}_1, \mathbf{e}_2) \neq 0$. In particular, $\mathbf{e}_1$ and $\mathbf{e}_2$ are linearly independent. Dividing $\mathbf{e}_1$ by a scalar, $\omega(\mathbf{e}_1, \mathbf{e}_2) = 1$, so the matrix of ω with respect to $\{\mathbf{e}_1, \mathbf{e}_2\}$ is

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Now assume the result true for dimension $< 2n$. Proceeding as before, find two vectors $\mathbf{e}_1, \mathbf{e}_{n+1}$ such that $\omega(\mathbf{e}_1, \mathbf{e}_{n+1}) = 1$. Let P be the plane spanned by these two vectors and consider the orthogonal $P^\perp$. It is immediate to check that $P \oplus P^\perp = E$. By the induction hypothesis, there exists a symplectic basis $\mathbf{e}_2, \dots, \mathbf{e}_n, \mathbf{e}_{n+2}, \dots, \mathbf{e}_{2n}$ of $P^\perp$. It is clear that $\mathbf{e}_1, \dots, \mathbf{e}_{2n}$ is a symplectic basis of E .

Finally, it follows from the matrix expression (3.1), that in any symplectic basis we have $\omega = \sum_{i=1}^n \alpha^i \wedge \alpha^{i+n}$. $\square$

Definition 3.1.6. A *symplectic isomorphism* between two symplectic vector spaces (E_1, ω_1) and (E_2, ω_2) is a linear isomorphism $\phi : E_1 \rightarrow E_2$ such that

$$\omega_1(\mathbf{x}, \mathbf{y}) = \omega_2(\phi(\mathbf{x}), \phi(\mathbf{y})), \quad \mathbf{x}, \mathbf{y} \in E_1,$$

or, equivalently, in terms of the pullback, $\omega_1 = \phi^* \omega_2$.

If A is the matrix of a linear map $\phi : E_1 \rightarrow E_2$ with respect to two symplectic bases, then the condition to be a symplectic isomorphism is

$$A^\top J A = J.$$

3.2 Symplectic manifolds

Definition 3.2.1. Let M be a smooth manifold. A *symplectic form* on M is a closed, non-degenerate 2-form $\omega \in \Omega^2(M)$. The pair (M, ω) is called a *symplectic manifold*.

By a non-degenerate 2-form on M we mean a differential 2-form ω such that for every $p \in M$ the induced linear 2-form ω_p is non-degenerate. In other words, ω_p is a symplectic inner product on the tangent space $T_p M$. Therefore, the dimension of $T_p M$, which is the same as that of M , must be even. We will write $\dim M = 2n$.

Example 3.2.2. Consider the usual coordinates $x^1, \dots, x^n, y_1, \dots, y_n$ on $\mathbb{R}^{2n}$. One can easily check that $\omega = dx^i \wedge dy_i$ is a symplectic form. Indeed, for each $p \in \mathbb{R}^{2n}$ the matrix of ω with respect to the basis $\{\frac{\partial}{\partial x^i}\}$ is precisely

$$\begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix}.$$

Every symplectic form can be expressed locally (and not only point-wise) as in the previous example:

Theorem 3.2.3 (Darboux). Let (M, ω) be a symplectic manifold. For every $p \in M$ there exists a local chart (U, x^i, y_i) such that $p \in U$ and

$$\omega|_U = dx^i \wedge dy_i.$$

Proof. We refer the reader to [3], where two proofs can be found. The first one is elementary but tedious, while the second one uses the so-called *Moser trick*, which makes it much shorter. $\square$

When we express ω as in the previous theorem, we say that we are using *Darboux (or canonical) coordinates*. Note that the last theorem implies that symplectic manifolds have no local invariants, as locally they are all the same.

It follows from the non-degeneracy of ω (it is enough to show it in Darboux coordinates), that the $2n$ -form

$$\omega^n = \omega \wedge \dots \wedge \omega$$

does not vanish, i.e., it is a volume form. It is sometimes called the *Liouville volume form* or *symplectic volume form*.

Remark 3.2.4. Some authors define the symplectic volume form of M as

$$\Omega = \frac{(-1)^{\lfloor n/2 \rfloor}}{n!} \omega^n,$$

where $\lfloor \frac{n}{2} \rfloor$ is the integer part of $\frac{n}{2}$, because this way it can be seen that the local expression of Ω in Darboux coordinates (U, x^i, y_i) is exactly

$$\Omega|_U = dx^1 \wedge \dots \wedge dx^n \wedge dy_1 \wedge \dots \wedge dy_n.$$

The existence of a volume form has a number of interesting consequences.

Proposition 3.2.5. *Let (M, ω) be a symplectic manifold. If M is compact, then ω is not exact.*

Proof. By way of contradiction, assume that ω is exact. Recall that the wedge product of exact forms is exact: $d\alpha \wedge d\beta = d(\alpha \wedge d\beta)$. By induction, ω^n is exact. Then, by Stoke's theorem,

$$\int_M \omega^n = 0,$$

which is absurd because ω^n is a volume form. □

The existence of a volume form is equivalent to the manifold M being oriented. Therefore:

Proposition 3.2.6. *Every symplectic manifold is oriented.*

In relation with the above proposition, we have the following example.

Example 3.2.7. Let us show that every oriented surface S is a symplectic manifold. Since S is oriented, it admits an area form ω . To check that the closed 2-form ω is a symplectic form, we need to see that ω_p is non-degenerate for every $p \in S$. Indeed, ω_p is represented with respect to any basis of $T_p M$ by a 2×2 skew-symmetric matrix

$$A = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}.$$

Since ω does not vanish, ω_p is not the null form, so $a \neq 0$. Therefore, $\det A = a^2 \neq 0$, meaning that ω_p is non-degenerate.

As a specific case of the previous example, we have:

Example 3.2.8. Consider the 2-sphere $\mathbb{S}^2$ with cylindrical coordinates (U, θ, h) (where U is an open set not including the poles; for example, the one given by $0 < \theta < 2\pi$ and $-1 < h < 1$). The usual area form is then

$$\omega|_U = d\theta \wedge dh.$$

Notice that these are Darboux coordinates for the symplectic form ω .

There is one more example of a symplectic manifold that will be crucial in the sequel.

Example 3.2.9. Let M be a smooth manifold of dimension m . We will see that the cotangent bundle T^*M has a natural symplectic structure. First, let us define a 1-form $\theta \in \Omega^1(T^*M)$ by the local expression

$$\theta|_U = y_i dx^i,$$

for any chart (U, x^i, y_i) of natural coordinates on T^*M (i.e., the coordinates induced from a coordinate system x^i on an open subset of M). The 1-form θ is called the *tautological 1-form* or *Liouville 1-form* of T^*M . To verify that it is well defined, let $(V, \bar{x}^i, \bar{y}_i)$ be another set of natural coordinates on T^*M with $U \cap V \neq \emptyset$. Recall that by the properties of the cotangent bundle, we have that

$$\bar{y}_j = y_i \frac{\partial x^i}{\partial \bar{x}^j}, \quad j = 1, \dots, m.$$

Therefore, using that $\frac{\partial x^i}{\partial \bar{y}^j} = 0$, on $U \cap V$ we have

$$y_i dx^i = y_i \frac{\partial x^i}{\partial \bar{x}^j} d\bar{x}^j = \bar{y}_j d\bar{x}^j,$$

which proves that θ is well defined. The 2-form given by

$$\omega = -d\theta,$$

is always a symplectic form. Indeed, it is clearly closed (actually, it is exact) and, in natural coordinates, it is given by

$$\omega|_U = dx^i \wedge dy_i,$$

which shows that ω is a symplectic form, since these are already Darboux coordinates for ω . We conclude that (T^*M, ω) is a symplectic manifold, and ω is called the *canonical 2-form* of the cotangent bundle of M .

Definition 3.2.10. Let N be a submanifold of a symplectic manifold (M, ω) . Then, we say that N is *isotropic* (respectively *Lagrangian* or *symplectic*) if for all $p \in N$, the tangent space $T_p N$ is an isotropic (respectively Lagrangian or symplectic) subspace of $T_p M$.

Equivalently, N is isotropic if and only if $\omega|_N = 0$. If, moreover, $\dim N = \frac{1}{2} \dim M$, then it is Lagrangian.

Example 3.2.11. Let us give an example of a Lagrangian submanifold, let f be a smooth function on $\mathbb{R}^n$. Then the graph N of the map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ given by

$$y_1 = \frac{\partial f}{\partial x^1}, \quad \dots, \quad y_n = \frac{\partial f}{\partial x^n}$$

is a Lagrangian submanifold of $(\mathbb{R}^{2n}, \omega)$, with ω the 2-form from Example 3.2.2, as

$$\omega|_N = dx^i \wedge dy_i = \frac{\partial^2 f}{\partial x^i \partial x^j} dx^i \wedge dx^j = \sum_{i < j} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \frac{\partial^2 f}{\partial x^j \partial x^i} \right) dx^i \wedge dx^j = 0.$$

Example 3.2.12. Let M be a smooth manifold and consider its cotangent bundle T^*M with its usual symplectic structure. Then, the manifold M is a Lagrangian submanifold of T^*M . If we identify M with its image under the zero section

$$\iota: M \rightarrow T^*M, \quad p \mapsto (p, 0).$$

In natural coordinates (x^i, y_i) , the image of the zero section is given by $y_i = 0$. Consequently, the pullback vanishes, $\iota^* dy_i = 0$. Since $\omega = dx^i \wedge dy_i$, it follows that $\iota^* \omega = 0$.

More generally, for any closed 1-form $\alpha \in \Omega^1(M)$, the image of α , namely

$$\text{Im } \alpha = \{(p, \alpha_p) : p \in M\} \subset T^*M,$$

is a submanifold of T^*M diffeomorphic to M . By the Poincaré lemma, for any $p \in M$ there is an open neighborhood U and a function $f \in \mathcal{C}^\infty(U)$ such that $\alpha|_U = df$. Taking coordinates x^i on U (after reducing U if necessary), the implicit equations of this submanifold on the corresponding natural coordinates (x^i, y_i) are

$$y_i = \frac{\partial f}{\partial x^i}.$$

By the argument of the previous example, it is a Lagrangian submanifold of T^*M .

On the other hand, the fibers of the cotangent bundle, that is, the preimages $\pi^{-1}(p) \simeq T_p^*M$ with $p \in M$ (where $\pi : T^*M \rightarrow M$ is the canonical projection), are also Lagrangian submanifolds of T^*M . In this case, the implicit equations of $\pi^{-1}(p)$ in natural coordinates are

$$x^i = x_0^i,$$

where x_0^i are the coordinates of p , so $dx^i|_{\pi^{-1}(p)} = 0$, which implies that $\omega|_{\pi^{-1}(p)} = 0$.

Finally, the notion of equivalence in the context of symplectic geometry is that of *symplectomorphic manifolds*.

Definition 3.2.13. Let (M_1, ω_1) and (M_2, ω_2) be two symplectic manifolds. A *symplectomorphism* between M_1 and M_2 is a diffeomorphism $\phi : M_1 \rightarrow M_2$ such that $\phi^*\omega_2 = \omega_1$.

Remark 3.2.14. Since the pullback commutes with the wedge product, symplectomorphisms also preserve the symplectic volume form: $\phi^*\Omega_2 = \Omega_1$.

3.3 Canonical isomorphism

It is a well-known fact from linear algebra that a non-degenerate bilinear form ω on a finite-dimensional real vector space E defines a linear map

$$\widehat{\omega} : E \rightarrow E^*,$$

which is an isomorphism. For every $v \in E$ the linear form $\widehat{\omega}(v)$ is defined by $\widehat{\omega}(v)(w) = \omega(v, w)$ for all $w \in E$. This carries over to a symplectic manifold (M, ω) naturally. Hence, we have the diffeomorphism

$$\widehat{\omega} : TM \rightarrow T^*M, \quad (p, v) \mapsto (p, \widehat{\omega}_p(v)).$$

In addition, this map extends to a $\mathcal{C}^\infty(M)$ -module isomorphism between vector fields and 1-forms, that assigns to each $X \in \mathfrak{X}(M)$ the form α defined by $\alpha(Y) = \omega(X, Y)$ for $Y \in \mathfrak{X}(M)$. This can be expressed in terms of the interior product as

$$\widehat{\omega} : \mathfrak{X}(M) \rightarrow \Omega^1(M), \quad X \mapsto \iota_X \omega.$$

The map $\widehat{\omega}$ is called the *canonical isomorphism* induced by ω .

Remark 3.3.1. In the previous comments there is nothing special about using a symplectic form. In fact, any covariant 2-tensor that is point-wise non-degenerate induces a canonical isomorphism between vector fields and 1-forms. For example, the metric tensor of a Riemannian (or pseudo-Riemannian) manifold also induces such an isomorphism.

The canonical isomorphism allows us to associate to each function $f \in \mathcal{C}^\infty(M)$ a unique vector field $X_f \in \mathfrak{X}(M)$.

Definition 3.3.2. Let (M, ω) be a symplectic manifold and let $f \in \mathcal{C}^\infty(M)$ be a function. The *Hamiltonian vector field* of f is $X_f = \widehat{\omega}^{-1}(df)$. Equivalently, X_f is the only vector field satisfying

$$\iota_{X_f} \omega = df.$$

A triplet (M, ω, f) is called a *Hamiltonian system*.

Remark 3.3.3. In the case of a canonical isomorphism induced by a Riemannian metric, the corresponding notion is that of the gradient vector field.

The importance of Hamiltonian fields is that, expressing the equations on Darboux coordinates, we obtain Hamilton's equations from mechanics, with f playing the role of the Hamiltonian function. Let us see this. If (M, ω) is a symplectic manifold, (U, x^i, y_i) is a local chart of Darboux coordinates, and we write $X_f|_U = A^i \frac{\partial}{\partial x^i} + B_i \frac{\partial}{\partial y_i}$, then

$$\iota_{X_f}\omega|_U = \iota_{X_f}(dx^i \wedge dy_i) = (\iota_{X_f}dx^i) \wedge dy_i - dx^i \wedge (\iota_{X_f}dy_i) = A^i dy_i - B_i dx^i.$$

On the other hand,

$$df|_U = \frac{\partial f}{\partial x^i} dx^i + \frac{\partial f}{\partial y_i} dy_i,$$

so the local expression of the Hamiltonian field of f is

$$X_f|_U = \frac{\partial f}{\partial y_i} \frac{\partial}{\partial x^i} - \frac{\partial f}{\partial x^i} \frac{\partial}{\partial y_i}. \quad (3.2)$$

Therefore, writing out the differential equations for the integral curves of X , we obtain Hamilton's equations:

$$\begin{cases} \dot{x}^i = \frac{\partial f}{\partial y_i} \\ \dot{y}_i = -\frac{\partial f}{\partial x^i} \end{cases}$$

(where $(x^1(t), \dots, x^n(t), y_1(t), \dots, y_n(t))$ is the local expression of an integral curve of X).

Note, however, that not every vector field X satisfies $\iota_X\omega = df$ for some function f , i.e. not every vector field is the Hamiltonian vector field for some function. For example, the form $\iota_X\omega$ might not be exact. In general, $\iota_X\omega$ is not even closed. This motivates the following definitions.

Definition 3.3.4. Let (M, ω) be a symplectic manifold. A vector field X is a *Hamiltonian vector field* if the form $\iota_X\omega$ is exact. A function $f \in \mathcal{C}^\infty(M)$ satisfying $\iota_X\omega = df$ is called a *Hamiltonian function* of X .

Definition 3.3.5. Let (M, ω) be a symplectic manifold. A vector field X is a *local Hamiltonian vector field* if the form $\iota_X\omega$ is closed.

In this case, for every $p \in M$, the Poincaré lemma ensures the existence of a neighborhood U and a function $f \in \mathcal{C}^\infty(U)$ such that $p \in U$ and $\iota_X\omega|_U = df$. Any function f satisfying this condition is called a *local Hamiltonian function* of X .

Remark 3.3.6. Obviously, every Hamiltonian field is locally Hamiltonian.

Remark 3.3.7. By the Cartan formula,

$$\mathcal{L}_X\omega = d\iota_X\omega + \iota_Xd\omega.$$

Since $d\omega = 0$, it is clear that X is locally Hamiltonian if and only if $\mathcal{L}_X\omega = 0$.

3.4 Poisson manifolds

Definition 3.4.1. Let (M, ω) be a symplectic manifold. The so-called *Poisson bracket* is the bilinear map

$$\{\cdot, \cdot\}: \mathcal{C}^\infty(M) \times \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M), \quad (f, g) \mapsto \{f, g\}$$

given by the formula $\{f, g\} = \omega(X_f, X_g)$. The function $\{f, g\}$ is called the *Poisson bracket* of f and g .

Proposition 3.4.2. Let (M, ω) be a symplectic manifold. The Poisson bracket satisfies the following properties:

1. $\{f, g\} = -\{g, f\}$ (skew symmetry)
2. $\{f, g\} = X_g f$
3. $\{f, gh\} = h\{f, g\} + g\{f, h\}$ (Leibniz rule)
4. $X_{\{f, g\}} = [X_g, X_f]$
5. $\{f, \{g, h\}\} + \{h, \{f, g\}\} + \{g, \{h, f\}\} = 0$ (Jacobi identity)

Proof.

1. Immediate by the skew-symmetry of ω .

2. We have that

$$\{f, g\} = \omega(X_f, X_g) = (\iota_{X_f} \omega)(X_g) = df(X_g) = X_g f.$$

3. Using the last property and the Leibniz rule for fields acting on functions,

$$\begin{aligned} \{f, gh\} &= -\{gh, f\} = -X_f(gh) = -gX_f h - hX_f g \\ &= -g\{h, f\} - h\{g, f\} = g\{f, h\} + h\{f, g\}. \end{aligned}$$

4. Using the formula $\iota_{[X, Y]} = \mathcal{L}_X \iota_Y - \iota_Y \mathcal{L}_X$ together with the Cartan formula,

$$\begin{aligned} \iota_{[X_g, X_f]} \omega &= \mathcal{L}_{X_g} \iota_{X_f} \omega - \iota_{X_f} \mathcal{L}_{X_g} \omega \\ &= d\iota_{X_g} \iota_{X_f} \omega + \iota_{X_g} d\iota_{X_f} \omega - \iota_{X_f} d\iota_{X_g} \omega - \iota_{X_f} \iota_{X_g} d\omega \\ &= d\omega(X_f, X_g) + \iota_{X_g} d^2 f - \iota_{X_f} d^2 g \\ &= d\{f, g\}, \end{aligned}$$

and the result follows from the definition of Hamiltonian field.

5. Since ω is closed,

$$\begin{aligned} 0 &= d\omega(X_f, X_g, X_h) \\ &= X_f \omega(X_g, X_h) - X_g \omega(X_f, X_h) + X_h \omega(X_f, X_g) \\ &\quad - \omega([X_f, X_g], X_h) + \omega([X_f, X_h], X_g) - \omega([X_g, X_h], X_f) \\ &= X_f \{g, h\} - X_g \{f, h\} + X_h \{f, g\} - \omega(X_{\{g, f\}}, X_h) + \omega(X_{\{h, f\}}, X_g) - \omega(X_{\{h, g\}}, X_f) \\ &= \{\{g, h\}, f\} - \{\{f, h\}, g\} + \{\{f, g\}, h\} - \{\{g, f\}, h\} + \{\{h, f\}, g\} - \{\{h, g\}, f\} \\ &= 2(\{\{g, h\}, f\} + \{\{h, f\}, g\} + \{\{f, g\}, h\}). \end{aligned}$$

The result follows from dividing this last expression by 2 and changing the sign.

□

By the local expression for Hamiltonian fields in Darboux coordinates (Eq. (3.2)) and the equality $\{f, g\} = X_g f$ it is immediate that the local expression of $\{f, g\}$ in Darboux coordinates (U, x^i, y_i) is

$$\{f, g\}|_U = \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial y_i} - \frac{\partial f}{\partial y_i} \frac{\partial g}{\partial x^i}.$$

In particular,

$$\{x^i, x^j\} = 0, \quad \{y_i, y_j\} = 0, \quad \{x^i, y_j\} = \delta_j^i.$$

Properties 1 and 5 of the previous proposition imply that $\mathcal{C}^\infty(M)$ together with the Poisson bracket form a real Lie algebra.

In general, given a manifold M , a bilinear map

$$\{\cdot, \cdot\}: \mathcal{C}^\infty(M) \times \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M), \quad (f, g) \mapsto \{f, g\}$$

satisfying properties 1, 3, and 5 of Proposition 3.4.2 is called a Poisson bracket on M . A pair $(M, \{\cdot, \cdot\})$ is called a Poisson manifold.

Remark 3.4.3. Not every Poisson bracket arises from a symplectic form. In fact, we can define a Poisson bracket on any manifold setting $\{f, g\} = 0$ for all functions f, g (but not every manifold admits a symplectic structure).

Example 3.4.4. A Poisson bracket on $\mathbb{R}^2$ can be defined by the formula

$$\{f, g\} = h \left(\frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} \right)$$

where $h \in \mathcal{C}^\infty(\mathbb{R}^2)$ is any function. If $h = 1$, it is just the Poisson bracket induced by the standard symplectic structure on $\mathbb{R}^2$.

Remark 3.4.5. Given a $(2, 0)$ -tensor field A on a manifold, the formula

$$\{f, g\} = A(df, dg)$$

defines a bilinear map over $\mathcal{C}^\infty(M)$. If the local expression of A with respect to a chart (U, x^i) is

$$A|_U = A^{ij} \frac{\partial}{\partial x^i} \otimes \frac{\partial}{\partial x^j},$$

then

$$\{f, g\}|_U = A^{ij} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j}.$$

It can be seen that for this to be a Poisson bracket we need A to be skew-symmetric (i.e., a bivector field) and that, for fixed i, j, k , it satisfies the relation

$$A^{il} \frac{\partial A^{jk}}{\partial x^l} + A^{jl} \frac{\partial A^{ki}}{\partial x^l} + A^{kl} \frac{\partial A^{ij}}{\partial x^l} = 0$$

corresponding to Jacobi identity. In fact, every Poisson bracket has an associated bivector field.

If A is non-degenerate (i.e., $\det(A^{ij}) \neq 0$), then M is a symplectic manifold with respect to the form ω whose components ω_{ij} result from inverting the matrix (A^{ij}) , that is, they are determined by $\omega_{ik} A^{kj} = \delta_i^j$ (and the Poisson bracket induced by ω is precisely $\{\cdot, \cdot\}$).

3.5 Symmetries of Hamiltonian systems

The trajectories of a Hamiltonian system (M, ω, h) are the solutions to the ordinary differential equation corresponding to the field X_h :

$$\frac{dc}{dt}(t) = X_h(c(t)),$$

where $c: I \rightarrow M$. Recall that the variation of a function $f \in \mathcal{C}^\infty(M)$ along an integral curve of a field X is

$$\frac{d(f \circ c)}{dt}(t) = X(f) \circ c(t),$$

or $(f \circ c)' = X(f) \circ c$ for short. In the case of the Hamiltonian field X_h , we have $X_h f = \{f, h\}$, so

$$\frac{d(f \circ c)}{dt} = \{f, h\} \circ c.$$

As with any differential equation, the *constants of motion* (also called *conserved quantities*) are the functions $f \in \mathcal{C}^\infty(M)$ that are constant along every integral curve c , i.e.,

$$\frac{d(f \circ c)}{dt} = 0.$$

For a general field X , a function f is a constant of motion if and only if $X(f) = 0$. In the Hamiltonian case, the constants of motion can be characterized in terms of the Poisson bracket by the equation $\{f, h\} = 0$.

Remarkably, the Hamiltonian function itself is conserved since $\{h, h\} = 0$. Thus, we have proved:

Theorem 3.5.1 (Conservation of energy). *In a Hamiltonian system (M, ω, h) , the Hamiltonian function h is a constant of motion.*

The following definition will play a very important role in the next chapter.

Definition 3.5.2. Let (M, ω) be a symplectic manifold. A collection of functions $f_1, \dots, f_r \in \mathcal{C}^\infty(M)$ are said to be *in involution* if $\{f_i, f_j\} = 0$ for all $1 \leq i, j \leq r$.

In other words, $f_1, \dots, f_r$ are in involution if each f_i is a constant of motion of all the Hamiltonian systems (M, ω, f_j) with $1 \leq j \leq r$.

Another immediate result concerning constants of motion is the following.

Theorem 3.5.3 (Poisson's theorem). *If f and g are constants of motion of a Hamiltonian system, then $\{f, g\}$ is also a constant of motion.*

Proof. This follows from the Jacobi identity, as

$$\{\{f, g\}, h\} = \{\{f, h\}, g\} + \{f, \{g, h\}\} = \{0, g\} + \{f, 0\} = 0.$$

□

4. Liouville–Arnold theorem

In 1853, J. Liouville (1809-1882) proved that, given a Hamiltonian system with n degrees of freedom, if there exist n constants of motion that are independent in a certain sense, then the equations of motion can be solved by quadratures, that is, they can be expressed explicitly in terms of integrals. This result was extended and popularized by V. I. Arnold (1937-2010) in his famous work *Mathematical Methods of Classical Mechanics* [2], and is thus known as the Liouville–Arnold theorem. In the first section, we introduce the notion of integrable system, state the Liouville–Arnold theorem and prove the first 3 parts, which are related to the so-called Liouville tori of the system. In the second section, the remaining parts, regarding action-angle variables, are proven. The last section is devoted to illustrate the theorem with various examples. The proof presented here is mainly based on [1, 2, 3]. Also, we use some intermediate results from Appendix A.

4.1 Integrable systems and Liouville tori

Definition 4.1.1. Let (M, ω, h) be a Hamiltonian system with $\dim(M) = 2n$. The system is called *integrable* if there exist n constants of motion $f_1, \dots, f_n$, called *integrals*, such that:

1. The functions $f_1, \dots, f_n$ are in involution.
2. The functions $f_1, \dots, f_n$ are *functionally independent*, i.e., $df_1 \wedge \dots \wedge df_n \neq 0$ almost everywhere.

The second condition means that the open set $\{p \in M : dpf_1 \wedge \dots \wedge dpf_n \neq 0\}$ is dense in M . In turn, the inequality

$$dpf_1 \wedge \dots \wedge dpf_n \neq 0$$

means that the dpf_i are linearly independent elements of T_p^*M .

Remark 4.1.2. It is impossible to have more than n functionally independent functions on M because if $r > n$, then $df_1 \wedge \dots \wedge df_r = 0$.

Remark 4.1.3. If $n = 1$, we can take $f_1 = h$, since the Hamiltonian is always a constant of motion. The first condition is trivially satisfied and the second one is reduced to $dh \neq 0$ almost everywhere. This also holds except in very pathological cases, so we can say that virtually any Hamiltonian system with one degree of freedom is integrable.

Example 4.1.4. Consider an n -dimensional harmonic oscillator, that is, a particle in $Q = \mathbb{R}^n$ subject to a central force $\mathbf{F} = -k\mathbf{x}$. The phase space is $T^*Q = \mathbb{R}^{2n}$ and the Hamiltonian turns out to be

$$h = \frac{1}{2} \sum_{i=1}^n \left(\frac{p_i^2}{m} + k(q^i)^2 \right),$$

usually written in terms of the angular frequency $\omega_0 = \sqrt{k/m}$ as

$$h = \frac{1}{2} \sum_{i=1}^n \left(\frac{p_i^2}{m} + m\omega_0^2(q^i)^2 \right).$$

To see that this is an integrable system, let us show that each term of the sum is an integral, i.e., if we take

$$f_i = \frac{1}{2} \left(\frac{p_i^2}{m} + m\omega_0^2(q^i)^2 \right),$$

then it is clear from the expression

$$\{f_i, f_j\} = \frac{\partial f_i}{\partial q^k} \frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \frac{\partial f_j}{\partial q^k}$$

that they are in involution because f_i only depends on q^i and p_i . Besides, they are constants of motion because

$$\{f_i, h\} = \left\{ f_i, \sum_{j=1}^n f_j \right\} = \sum_{j=1}^n \{f_i, f_j\} = 0.$$

Finally, the forms

$$df_i = m\omega_0^2 q_i dq_i + \frac{p_i}{m} dp_i$$

are linearly independent almost everywhere (as they only vanish if $q^i = p_i = 0$).

Recall that given two smooth manifolds M and N , of dimensions m and n respectively, and a smooth map $F : M \rightarrow N$, a *regular value* of F is a point $q \in N$ such that for every $p \in F^{-1}(q)$ the tangent map $T_p F : T_p M \rightarrow T_q N$ is a surjection. Under these conditions, the well-known regular value theorem states that $P = F^{-1}(q)$ is a submanifold of M of dimension $m - n$.

Given an integrable system (M, ω, h) with integrals $f_1, \dots, f_n$, we can construct the smooth map

$$F = (f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n.$$

For any regular value $\Lambda \in \mathbb{R}^n$ of the previous map, we can consider the submanifold

$$M_\Lambda = \{p \in M : f_i(p) = \Lambda_i, \quad i = 1, \dots, n\},$$

of dimension n . We are now ready to state the Liouville–Arnold theorem.

Theorem 4.1.5 (Liouville–Arnold). *Let (M, ω, h) be an integrable system with integrals $f_1, \dots, f_n$. Let $\Lambda \in \mathbb{R}^n$ be a regular value of the map*

$$(f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n.$$

Consider the level set

$$M_\Lambda = \{p \in M : f_i(p) = \Lambda_i, \quad i = 1, \dots, n\}.$$

Then:

1. M_Λ is a Lagrangian submanifold of (M, ω) .
2. M_Λ is invariant with respect to the flow of the X_{f_i} and X_h .
3. Any compact connected component of M_Λ is diffeomorphic to the n -dimensional torus $\mathbb{T}^n$. Any such component is called a Liouville torus of the Hamiltonian system.
4. Any compact connected component of M_Λ has an open neighborhood U that is diffeomorphic to $\mathbb{T}^n \times \mathbb{D}^n$.

5. On U , there are coordinates (φ^i, s_i) (with φ^i periodic), called action-angle variables, verifying the following properties.

- (a) They are Darboux coordinates for ω , i.e., $\omega|_U = d\varphi^i \wedge ds_i$.
- (b) The action variables s_i are functions of the integrals $f_1, \dots, f_n$, so they are constants of motion.
- (c) The integral curves of X_h are given by

$$\dot{\varphi}^i = q^i(s_1, \dots, s_n), \quad \dot{s}_i = 0.$$

Broadly speaking, the first part of the theorem establishes that the regular level sets M_Λ are Lagrangian submanifolds invariant under the joint Hamiltonian flows. The remaining parts characterize the rigid topology of these sets: when a connected component is compact, it is necessarily diffeomorphic to an n -dimensional torus $\mathbb{T}^n$, around which the dynamics can be completely linearized using action-angle variables.

In this section, we will prove the first 3 parts of the theorem.

Proof of part 1. Fix $p \in M_\Lambda$. Since Λ is a regular value, $T_p F : T_p M \rightarrow \mathbb{R}^n$ is exhaustive. Notice that $T_p F = (d_p f_1, \dots, d_p f_n)$. This linear map being full rank (due to exhaustivity) means that $d_p f_1, \dots, d_p f_n$ are linearly independent. By the canonical isomorphism, $X_{f_1}(p), \dots, X_{f_n}(p)$ are also linearly independent. Since they are tangent to M_Λ , they form a basis of $T_p M_\Lambda$. Now,

$$\omega(X_{f_i}, X_{f_j}) = \{f_i, f_j\} = 0, \quad 1 \leq i, j \leq n,$$

so $\omega_p(X_{f_i}(p), X_{f_j}(p)) = 0$. By linearity, $\omega_p|_{T_p M_\Lambda} = 0$. Since p was arbitrary, we conclude that $\omega|_{M_\Lambda} = 0$. $\square$

Proof of part 2. Each f_j is a constant of motion with respect to X_{f_i} and X_h , since

$$X_{f_i} f_j = \{f_j, f_i\} = 0, \quad X_h f_j = \{f_j, h\} = 0.$$

Therefore, the sets

$$M_{\Lambda_j} = \{p \in M : f_j(p) = \Lambda_j\}$$

are invariant with respect to X_{f_i} and X_h . Finally, M_Λ is also invariant as it is an intersection of invariant sets:

$$M_\Lambda = \bigcap_{j=1}^n M_{\Lambda_j}$$

$\square$

In the proof of part 1 we have also seen that the restrictions of $X_{f_1}, \dots, X_{f_n}$ to M_Λ are pointwise linearly independent vector fields, i.e., they are a smooth frame on M_Λ . Besides, they are pairwise commutative vector fields, since

$$[X_{f_i}, X_{f_j}] = X_{\{f_j, f_i\}} = 0.$$

This reminds of the n -dimensional torus

$$\mathbb{T}^n = \overbrace{\mathbb{S}^1 \times \dots \times \mathbb{S}^1}^n$$

where we have a smooth frame defined as follows. First, the local diffeomorphism

$$\mathbb{R} \rightarrow \mathbb{S}^1, \theta \mapsto (\cos \theta, \sin \theta)$$

can be generalized to a local diffeomorphism

$$\pi : \mathbb{R}^n \rightarrow \mathbb{T}^n, (\theta^1, \dots, \theta^n) \mapsto (\cos \theta^1, \sin \theta^1, \dots, \cos \theta^n, \sin \theta^n).$$

Restricting this map to sufficiently small open sets U gives rise to parametrizations of the torus. The corresponding charts (U, θ^i) are the usual angle coordinates of the torus. The angle coordinates of a point may differ by a multiple of 2π depending on the chart, but the partial derivatives

$$X_i = \frac{\partial}{\partial \theta^i}$$

are actually globally defined. The fields of the frame $\{X_i\}$ are pairwise commutative because of Schwarz's theorem:

$$\left[\frac{\partial}{\partial \theta^i}, \frac{\partial}{\partial \theta^j} \right] = 0$$

More generally, the following lemma shows that any compact connected manifold admitting such a smooth frame is necessarily diffeomorphic to a torus.

Lemma 4.1.6. *Let N be a connected n -dimensional manifold and let $X_1, \dots, X_n \in \mathfrak{X}(N)$ be a smooth frame of complete vector fields. If these vector fields are pairwise commutative, then N is diffeomorphic to $\mathbb{T}^k \times \mathbb{R}^{n-k}$ for some integer $0 \leq k \leq n$. In particular, if N is compact, then $N \simeq \mathbb{T}^n$.*

Proof. Let $\phi_i^t : N \rightarrow N$ be the flow of X_i in the instant t . The flow is defined for every $t \in \mathbb{R}$ because the fields are complete. Consider the map

$$\Phi : \mathbb{R}^n \times N \rightarrow N, \quad (\mathbf{t}, p) \mapsto (\phi_1^{t_1} \circ \dots \circ \phi_n^{t_n})(p).$$

Let us see that this is a group action of $\mathbb{R}^n$ on N . First, for each $\mathbf{t} \in \mathbb{R}^n$ the map

$$\Phi^{\mathbf{t}} : N \rightarrow N, \quad p \mapsto \Phi(\mathbf{t}, p)$$

is a diffeomorphism of N because it is a composition of diffeomorphisms: $\Phi^{\mathbf{t}} = \phi_1^{t_1} \circ \dots \circ \phi_n^{t_n}$. The commutativity of the vector fields X_i , i.e., $[X_i, X_j] = 0$ for all i, j , implies that the flows ϕ_i^t commute:

$$\phi_i^t \circ \phi_j^s = \phi_j^s \circ \phi_i^t, \quad t, s \in \mathbb{R}, 1 \leq i, j \leq n.$$

This, together with the group properties of the flow, implies that

$$\begin{aligned} \Phi^{\mathbf{t}+\mathbf{s}} &= \phi_1^{t_1+s_1} \circ \dots \circ \phi_n^{t_n+s_n} = \phi_1^{t_1} \circ \phi_1^{s_1} \circ \dots \circ \phi_n^{t_n} \circ \phi_n^{s_n} \\ &= \phi_1^{t_1} \circ \dots \circ \phi_n^{t_n} \circ \phi_1^{s_1} \circ \dots \circ \phi_n^{s_n} = \Phi^{\mathbf{t}} \circ \Phi^{\mathbf{s}} \end{aligned}$$

for $\mathbf{t}, \mathbf{s} \in \mathbb{R}^n$, so Φ is a group action.

Next, let us show that for every $p \in N$ the map

$$\Phi_p : \mathbb{R}^n \rightarrow N, \quad \mathbf{t} \mapsto \Phi(\mathbf{t}, p)$$

is a local diffeomorphism. Let $\mathbf{t} \in \mathbb{R}^n$ and denote $q = \Phi_p(\mathbf{t})$. We must show that the tangent map $T_{\mathbf{t}}\Phi_p : \mathbb{R}^n \rightarrow T_q N$ is an isomorphism. Let $\{\mathbf{e}_i\}$ denote the canonical basis of $\mathbb{R}^n$. We can compute the image of $\mathbf{e}_i$ as $T_{\mathbf{t}}\Phi_p(\mathbf{e}_i) = (\Phi_p \circ \alpha)'(0)$ where $\alpha(t) = \mathbf{t} + te_i$, a curve on $\mathbb{R}^n$ with $\alpha(0) = \mathbf{t}$ and $\alpha'(0) = \mathbf{e}_i$. Now, by the group properties of the flow,

$$(\Phi_p \circ \alpha)(t) = \Phi_p(te_i + \mathbf{t}) = \Phi^{te_i + \mathbf{t}}(p) = \Phi^{te_i}(\Phi^{\mathbf{t}}(p)) = \phi_i^t(q).$$

And hence,

$$(\Phi_p \circ \alpha)'(0) = \left. \frac{d}{dt} \right|_{t=0} (\phi_i^t(q)) = X_i(\phi_i^0(q)) = X_i(q).$$

Since the X_i form a smooth frame, $\{X_i(q)\}$ is a basis of $T_q N$. $T_{\mathbf{t}}\Phi_p$ sends a basis to a basis, and thus it is an isomorphism.

This establishes that Φ_p is a local diffeomorphism and, in particular, an open map. Thus, the image of Φ_p , which is precisely the orbit of p , is an open subset of N . Since p is arbitrary, every orbit in N is open. Because the orbits of a group action partition the manifold, the complement of any orbit is a union of open sets, hence closed. Therefore, each orbit is both open and closed. Since N is connected, the only non-empty clopen subset is N itself, implying there is only one orbit (the action is transitive). In other words, Φ_p is surjective.

We claim that the isotropy group G_p of p is a discrete subgroup of $\mathbb{R}^n$. Since Φ_p is a local diffeomorphism, for every $\mathbf{t} \in G_p$ there is a neighborhood U of p in $\mathbb{R}^n$ such that $\Phi_p|_U$ is one-to-one, so there is no other $\mathbf{s} \in U$ such that $\Phi(\mathbf{s}, p) = p$, i.e., $\mathbf{t}$ is an isolated point of G_p in $\mathbb{R}^n$. Therefore, G_p is a discrete subgroup of $\mathbb{R}^n$. However, the discrete subgroups of $\mathbb{R}^n$ are known to be *lattices*. This means that there exists an integer $0 \leq k \leq n$ and linearly independent vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ such that G_p consists of the *integral* linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_k$ (the case $k = 0$ corresponding to $G_p = \{0\}$). Explicitly,

$$G_p = \left\{ \sum_{i=1}^k a_i \mathbf{v}_i : a_1, \dots, a_k \in \mathbb{Z} \right\}.$$

In other words, as a group, $G_p \simeq \mathbb{Z}^k$.

Now let us consider the local diffeomorphism

$$\pi : \mathbb{R}^n \rightarrow \mathbb{T}^k \times \mathbb{R}^{n-k}, \quad (\theta^1, \dots, \theta^k, x^1, \dots, x^{n-k}) \mapsto (e^{i\theta^1}, \dots, e^{i\theta^k}, x^1, \dots, x^{n-k}).$$

The vectors $\mathbf{w}_i = 2\pi\mathbf{e}_i$ for $i = 1, \dots, k$ are linearly independent, so there exists a linear isomorphism $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $A(\mathbf{w}_i) = \mathbf{v}_i$ for $i = 1, \dots, k$. We will now show that there exists a diffeomorphism φ that closes the following diagram.

$$\begin{array}{ccc} \mathbb{R}^n & \xrightarrow{A} & \mathbb{R}^n \\ \pi \downarrow & & \downarrow \Phi_p \\ \mathbb{T}^k \times \mathbb{R}^{n-k} & \xrightarrow{\varphi} & N \end{array}$$

Since we want the diagram to commute, we must have $\varphi \circ \pi = \Phi_p \circ A$, so given $q \in \mathbb{T}^k \times \mathbb{R}^{n-k}$, we define $\varphi(q) = \Phi_p(A(\mathbf{x}))$ where $\mathbf{x} \in \mathbb{R}^n$ is such that $\pi(\mathbf{x}) = q$. We will start showing that φ is well defined and one-to-one. If $\pi(\mathbf{x}) = \pi(\mathbf{y})$, the last $n - k$ coordinates must be the same and the first k at most differ in an integer multiple of 2π . Thus, there are integers $a_1, \dots, a_k \in \mathbb{Z}$ such that

$$\mathbf{x} - \mathbf{y} = (2\pi a_1, \dots, 2\pi a_k, 0, \dots, 0) = \sum_{i=1}^k a_i \mathbf{w}_i.$$

By linearity,

$$A(\mathbf{x}) = A\left(\mathbf{y} + \sum_{i=1}^k a_i \mathbf{w}_i\right) = A(\mathbf{y}) + \sum_{i=1}^k a_i \mathbf{v}_i.$$

The last term $\mathbf{t} = \sum_{i=1}^k a_i \mathbf{v}_i$ belongs to G_p , so we have $\Phi^{\mathbf{t}}(p) = p$. Thus,

$$\Phi_p(A(\mathbf{x})) = \Phi^{A(\mathbf{x})}(p) = \Phi^{A(\mathbf{y})+\mathbf{t}}(p) = \Phi^{A(\mathbf{y})}(\Phi^{\mathbf{t}}(p)) = \Phi^{A(\mathbf{y})}(p) = \Phi_p(A(\mathbf{y})).$$

Therefore, φ is well defined. Conversely, the same argument shows that if $\Phi_p(A(\mathbf{x})) = \Phi_p(A(\mathbf{y}))$, then $\pi(\mathbf{x}) = \pi(\mathbf{y})$, which means that φ is one-to-one.

Finally, let us see that φ is a diffeomorphism. For the surjectivity, let $q \in N$. A and Φ_p are surjective, so there is vector $\mathbf{x} \in \mathbb{R}^n$ such that $\Phi_p(A(\mathbf{x})) = q$, i.e., $\varphi(\pi(\mathbf{x})) = q$. This establishes that φ is onto and thus bijective. On the other hand, since π is a surjective local diffeomorphism, every point in $\mathbb{T}^k \times \mathbb{R}^{n-k}$ has a neighborhood U such that π has a local inverse $\eta : U \rightarrow \mathbb{R}^n$. By definition,

$$\varphi|_U = \Phi_p \circ A \circ \eta|_U.$$

In addition to smoothness, this proves that φ is a local diffeomorphism. Since it is bijective, it must be a diffeomorphism. This completes the proof. $\square$

Remark 4.1.7. Recall that, in a compact manifold, every vector field is complete.

This last lemma allows us to prove part 3 of the Liouville–Arnold theorem.

Proof of part 3. Let N be a compact component of M_Λ (which is a submanifold of the same dimension). The restrictions of the commutative fields X_{f_i} to M_Λ form a smooth frame of M_Λ , so $\{X_{f_1}|_N, \dots, X_{f_n}|_N\}$ is a smooth frame of N . Since N is compact, these vector fields are complete. By Lemma 4.1.6, $N \simeq \mathbb{T}^n$. $\square$

4.2 Action-angle variables

In this section, we will prove the remaining parts of Liouville–Arnold theorem, which are the ones regarding action-angle variables, for this we will use some results proved in Appendix A. In particular, the last statement of part 5 justifies the name *integrable system*, because the equations of motion written in action-angle variables take the form

$$\begin{cases} \dot{\varphi}^i = q^i(s_1, \dots, s_n), \\ \dot{s}_i = 0, \end{cases}$$

and this system can be easily integrated. Indeed, it implies that each s_i is constant, and so is each $q^i(s_1, \dots, s_n)$, so the solution is:

$$\begin{cases} \varphi^i(t) = q^i(s_1(0), \dots, s_n(0))t + \varphi^i(0), \\ s_i(t) = s_i(0). \end{cases}$$

Furthermore, in the proof we will see that $q^i = \frac{\partial h}{\partial s_i}$, with $h \in \mathcal{C}^\infty(M)$ the Hamiltonian of the function (which happens to depend only on the s_i).

The angle coordinates φ are periodic coordinates, similar to those of a torus. A periodic coordinate system on a manifold M is a pair (U, θ^i) consisting of an open set U and functions $\theta^i : U \rightarrow \mathbb{R}/2\pi\mathbb{Z}$

(real numbers modulo 2π) for $i = 1, \dots, n$, with the property that we can cover U with ordinary coordinate charts (V, x^i) such that

$$x^i(p) = \theta^i(p) \pmod{2\pi}, \quad p \in U \cap V.$$

The x^i represent a smooth choice of representatives of the θ^i on $U \cap V$. It is also possible to have a chart where only some variables are periodic.

Although periodic coordinates are not ordinary coordinates, the two share many properties. For example, we can define coordinate vector fields $\frac{\partial}{\partial \theta^i}$. If the charts (V, x^i) and (V, y^i) are compatible with (U, θ^i) (i.e. they satisfy the previous equation), then there exists an integer-valued function k^i such that $y^i - x^i = 2\pi k^i$. Being a smooth function that only takes integer values, k^i must be locally constant, so

$$y^i = x^i + 2\pi k^i \implies \frac{\partial y^i}{\partial x^j} = \delta_j^i \implies \frac{\partial}{\partial x^i} = \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j} = \frac{\partial}{\partial y^i}.$$

This defines a field $\frac{\partial}{\partial \theta^i}$ on all of U . Similarly, the 1-form $d\theta^i$ is globally defined.

Proof of part 4. For simplicity, we will start assuming that M_Λ itself is compact and connected. By Theorem A.3, there exists an open neighborhood D around Λ in $\mathbb{R}^n$ and a diffeomorphism $M_\Lambda \times D \rightarrow F^{-1}(D)$. As noted in Remark A.6, we can assume that D is a disk, i.e., $D \simeq \mathbb{D}^n$. So $U = F^{-1}(D)$ is an open neighborhood of M_Λ with

$$U \simeq M_\Lambda \times D \simeq \mathbb{T}^n \times \mathbb{D}^n.$$

If M_Λ is not connected, we can take disjoint open sets around each compact component and apply Theorem A.3 to the restriction of F to each open set. $\square$

Proof of part 5. As in the previous part, we can assume that M_Λ is compact and connected. First, we will see how the proof of Lemma 4.1.6 provides a specific way to define angle variables on M_Λ . Fix a point $p \in M_\Lambda$ and consider the commutative diagram from Figure 1 (left), which we have from Lemma 4.1.6. Recall that Φ is defined as

$$\Phi(\mathbf{t}, p) = \phi_1^{t_1} \circ \dots \circ \phi_n^{t_n}(p),$$

where ϕ_i is the flow of the Hamiltonian field X_{f_i} , while A is a linear map determined by the condition $A(2\pi \mathbf{e}_i) = \mathbf{v}_i$ for $i = 1, \dots, n$, where $\{\mathbf{v}_i\}$ is the integral basis of the lattice G_p .

To describe angle variables it is more convenient to think of the torus as the quotient $\mathbb{R}^n / 2\pi\mathbb{Z}^n$ by means of the usual identification

$$\gamma : \mathbb{R}^n / 2\pi\mathbb{Z}^n \rightarrow \mathbb{T}^n, (\theta^1 \bmod 2\pi, \dots, \theta^n \bmod 2\pi) \mapsto (\cos \theta^1, \sin \theta^1, \dots, \cos \theta^n, \sin \theta^n).$$

Therefore, the diagram shown in Figure 2 also commutes, where by definition $\theta = \gamma^{-1} \circ \varphi^{-1}$ and the map $\mathbb{R}^n \rightarrow \mathbb{R}^n / 2\pi\mathbb{Z}^n$ is the canonical projection. Removing $\mathbb{T}^n$, the diagram looks like Figure 1 (right), so we have successfully replaced $\mathbb{T}^n$ with $\mathbb{R}^n / 2\pi\mathbb{Z}^n$. The diffeomorphism $\theta = (\theta^1, \dots, \theta^n)$ defines angle coordinates $\theta^i(q)$ for $q \in M_\Lambda$. Of course, these are *periodic* coordinates.

Unpacking the definitions of the involved maps, we will now explicitly describe how to compute angle coordinates of a point $q \in M_\Lambda$. First, choose $\mathbf{t} \in \mathbb{R}^n$ such that $\Phi(\mathbf{t}, p) = q$. Express $\mathbf{t} = \sum_{i=1}^n a_i \mathbf{v}_i$ with $a_i \in \mathbb{R}$. Then,

$$A^{-1}(\mathbf{t}) = \sum_{i=1}^n a_i A^{-1}(\mathbf{v}_i) = 2\pi \sum_{i=1}^n a_i \mathbf{e}_i = (2\pi a_1, \dots, 2\pi a_n),$$

$$\begin{array}{ccc}
 \mathbb{R}^n & \xrightarrow{A} & \mathbb{R}^n \\
 \downarrow \pi & & \downarrow \Phi_p \\
 \mathbb{T}^n & \xleftarrow{\varphi} & M_\Lambda
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathbb{R}^n & \xrightarrow{A} & \mathbb{R}^n \\
 \downarrow & & \downarrow \Phi_p \\
 \mathbb{R}^n/2\pi\mathbb{Z}^n & \xleftarrow{\theta} & M_\Lambda
 \end{array}$$

Figure 1: (Left) Commutative diagram that arises following the proof of Lemma 4.1.6. (Right) Result of replacing the torus $\mathbb{T}^n$ with the quotient $\mathbb{R}^n/2\pi\mathbb{Z}^n$ in the previous diagram.

$$\begin{array}{ccc}
 & \mathbb{R}^n & \xrightarrow{A} \mathbb{R}^n \\
 & \downarrow \pi & \downarrow \Phi_p \\
 & \mathbb{T}^n & \xrightarrow{\varphi} M_\Lambda \\
 \swarrow \gamma & & \searrow \theta \\
 \mathbb{R}^n/2\pi\mathbb{Z}^n & &
 \end{array}$$

Figure 2: Natural extension of the diagram in Figure 1 (left) after introducing the quotient $\mathbb{R}^n/2\pi\mathbb{Z}^n$.

so

$$\theta^i(q) = 2\pi a_i \pmod{2\pi}, \quad i = 1, \dots, n,$$

or $\theta(q) = A^{-1}(\mathbf{t}) \pmod{2\pi}$ for short. Note that these coordinates do not depend on the choice of $\mathbf{t}$ because the θ are well-defined.

One last remark about the coordinates θ^i . Let $X_{f_i} = A_i^j \frac{\partial}{\partial \theta^j}$ be the expression of X_{f_i} with respect to the angle variables. We claim that the A_i^j are constant. The differential equation $c'(t) = X_{f_i}(c(t))$ with $c(0) = p$ is written locally as

$$\dot{\theta}^j = A_i^j(\theta^1, \dots, \theta^n), \quad j = 1, \dots, n.$$

However, using the flow of X_{f_i} , we can write

$$c(t) = \phi_i^t(p) = \Phi(te_i, p),$$

so the vector of coordinates of $c(t)$ is

$$\theta(c(t)) = A^{-1}(te_i) = tA^{-1}(\mathbf{e}_i)$$

More explicitly, $\theta^j(t) = \alpha_i^j t$ where (α_i^j) is the matrix representation of A^{-1} , so

$$\dot{\theta}^j = \alpha_i^j$$

and the $A_i^j = \alpha_i^j$ are constant.

Moving forward, we observe that, by part 4, there exists an open neighborhood of M_Λ of the form $U \simeq \mathbb{T}^n \times \mathbb{D}^n$. This suggests that the angle variables might make sense in all of U . Next, we will extend the θ^i to U and show that (θ^i, f_i) are coordinates on U .

By Theorem A.3, there exists $\eta : U \rightarrow M_\Lambda$ such that $(\eta, F) : U \rightarrow M_\Lambda \times D$ is a diffeomorphism (see Remark A.5). It follows that the restriction of η to any fiber $F^{-1}(y)$ with $y \in D$ is a diffeomorphism onto M_Λ . Replacing η with $\tilde{\eta} = (\eta|_{M_\Lambda})^{-1} \circ \eta$, we can assume that η is the identity on M_Λ . Therefore, we can extend the definition of θ to the neighborhood U putting $\tilde{\theta} = \theta \circ \eta$, so $\tilde{\theta} : U \rightarrow \mathbb{R}^n/2\pi\mathbb{Z}^n$ and the angle coordinates θ^i are defined on all of U . The map

$$(\tilde{\theta}, F) : U \rightarrow \mathbb{R}^n/2\pi\mathbb{Z}^n \times D$$

is a diffeomorphism because $(\eta, F) : U \rightarrow M_\Lambda \times D$ and $\theta : M_\Lambda \rightarrow \mathbb{R}^n/2\pi\mathbb{Z}^n$ are also diffeomorphisms. Therefore, (U, θ^i, f_i) is a coordinate system (where the first n coordinates are periodic coordinates).

The idea of the proof is to apply successive transformations to (θ^i, f_i) until we arrive to the action-angle variables. The expression of ω with respect to (θ^i, f_i) is

$$\omega|_U = \sum_{i < j} a_{ij} d\theta^i \wedge d\theta^j + \sum_{i < j} b^{ij} df_i \wedge df_j + \sum_{i,j=1}^n c_i^j d\theta^i \wedge df_j$$

By part 1, ω vanishes on M_Λ . Since the f_i are constant on M_Λ , we have

$$0 = \omega|_{M_\Lambda} = \sum_{i < j} a_{ij} d\theta^i \wedge d\theta^j,$$

so the a_{ij} vanish on M_Λ . Actually, by the same reasoning, the a_{ij} vanish on each level set $F^{-1}(\Lambda')$ with $\Lambda' \in D$, because all of them are Lagrangian submanifolds. Since these level sets make up all of U , it follows that $a_{ij} = 0$.

Thus, we can write ω as follows. For convenience, we renamed some indices.

$$\omega|_U = \sum_{i < j} b^{ij} df_i \wedge df_j + \sum_{i,j=1}^n c_i^j d\theta^i \wedge df_j \quad (4.1)$$

Our next goal is to show that both the c_i^j and the b^{ij} depend only on the f_i (and not on the θ^i). Starting with the c_i^j , we have seen before that, in a fixed Liouville torus, the frames $\{X_{f_i}\}$ and $\{\frac{\partial}{\partial \theta^i}\}$ are connected by a linear change, that is, the coefficients of the expression $\frac{\partial}{\partial \theta^i} = \ell_i^j X_{f_j}$ are constant on the torus. This means that they do not depend on the angle variables, but only on the f_i . Let us see that actually $c_i^j = \ell_i^j$:

$$\begin{aligned} c_i^j &= \omega \left(\frac{\partial}{\partial \theta^i}, \frac{\partial}{\partial f_j} \right) = \omega \left(\ell_i^k X_{f_k}, \frac{\partial}{\partial f_j} \right) = \ell_i^k \omega \left(X_{f_k}, \frac{\partial}{\partial f_j} \right) \\ &= \ell_i^k df_k \left(\frac{\partial}{\partial f_j} \right) = \ell_i^k \frac{\partial f_k}{\partial f_j} = \ell_i^k \delta_k^j = \ell_i^j \end{aligned}$$

In order to show that the b^{ij} do not depend on the θ^i , we will use the fact that ω is closed. First of all, let us write

$$\omega|_U = \beta + d\theta^i \wedge \omega_i$$

where

$$\beta = \sum_{i < j} b^{ij} df_i \wedge df_j, \quad \omega_i = c_i^j df_j.$$

Thus,

$$0 = d\omega|_U = d\beta - d\theta^i \wedge d\omega_i \implies d\beta = d\theta^i \wedge d\omega_i.$$

Let us now compare both sides of the last equation.

$$\begin{aligned} d\beta &= \sum_{i < j} \left(\frac{\partial b^{ij}}{\partial \theta^k} d\theta^k \wedge df_i \wedge df_j + \frac{\partial b^{ij}}{\partial f_k} df_k \wedge df_i \wedge df_j \right) \\ d\omega_k &= d(c_k^j df_j) = \sum_{i < j} \left(\frac{\partial c_k^j}{\partial f_i} - \frac{\partial c_k^i}{\partial f_j} \right) df_i \wedge df_j \\ d\theta^k \wedge d\omega_k &= \sum_{i < j} \left(\frac{\partial c_k^j}{\partial f_i} - \frac{\partial c_k^i}{\partial f_j} \right) d\theta^k \wedge df_i \wedge df_j \end{aligned}$$

Since in both expressions the components $df_i \wedge df_j \wedge d\theta^k$ (with $i < j$) appear only once, we deduce that

$$\frac{\partial b^{ij}}{\partial \theta^k} = \frac{\partial c_k^j}{\partial f_i} - \frac{\partial c_k^i}{\partial f_j}.$$

We see that $\frac{\partial b^{ij}}{\partial \theta^k}$ does not depend on θ^k , so

$$b^{ij} = a\theta^k + b$$

where the functions a and b do not depend on θ^k either. Since b^{ij} is periodic with respect to θ^k , it must be $a = 0$, so b^{ij} itself does not depend on θ^k .

In particular,

$$\frac{\partial c_k^j}{\partial f_i} - \frac{\partial c_k^i}{\partial f_j} = 0,$$

so $d\omega_k = 0$. Thus,

$$d\beta = d\theta^k \wedge d\omega_k = 0,$$

so ω_i and β are closed. However, since they only depend on the f_i , these forms can be regarded as forms on the disk D . Since every closed form on a disk is exact, ω_i and β are exact. Therefore, we can write $\omega_i = ds_i$ for some functions $s_i = s_i(f_1, \dots, f_n)$, and $\beta = d\kappa$ for a 1-form κ (depending only on the f_i). The symplectic form is then

$$\omega|_U = d\kappa + d\theta^i \wedge ds_i.$$

It follows from Eq. (4.1) that the matrix of ω with respect to (θ^i, f_i) is

$$\Omega = \begin{pmatrix} 0 & c_i^j \\ -c_i^j & b^{ij} \end{pmatrix},$$

where the lower triangular part of the matrix (b^{ij}) is completed by skew-symmetry, i.e., $b^{ij} = -b^{ji}$. Since ω is non-degenerate,

$$0 \neq \det \Omega = \det(c_i^j)^2 \implies \det(c_i^j) \neq 0.$$

But $c_i^j = \frac{\partial s_i}{\partial f_j}$ because

$$c_i^j df_j = \omega_i = ds_i = \frac{\partial s_i}{\partial f_j} df_j,$$

so the Jacobian matrix of the transformation $(\theta^i, f_i) \mapsto (\theta^i, s_i)$ is

$$\begin{pmatrix} \frac{\partial \theta^i}{\partial \theta^j} & \frac{\partial \theta^i}{\partial f_j} \\ \frac{\partial s_i}{\partial \theta^j} & \frac{\partial s_i}{\partial f_j} \end{pmatrix} = \begin{pmatrix} \delta_{ij} & 0 \\ 0 & c_i^j \end{pmatrix}.$$

Since this matrix is non-singular, (θ^i, s_i) is a coordinate system on U .

Finally, we will do one more transformation to obtain Darboux coordinates. We will shift the angle coordinates to cancel the last term. Express $\kappa = g^i ds_i$ and define $\varphi^i = \theta^i + g^i(s_1, \dots, s_n)$. Since the s_i are functions of $f_1, \dots, f_n$, they are constant along each torus, so φ^i are also angle coordinates, only that we changed the initial points from which the angles are measured in each torus. The Jacobian matrix of the transformation $(\theta^i, s_i) \mapsto (\varphi^i, s_i)$ is

$$\begin{pmatrix} \frac{\partial \varphi^i}{\partial \theta^j} & \frac{\partial \varphi^i}{\partial s_j} \\ \frac{\partial s_i}{\partial \theta^j} & \frac{\partial s_i}{\partial s_j} \end{pmatrix} = \begin{pmatrix} \delta_j^i & -\frac{\partial g^i}{\partial s_j} \\ 0 & \delta_i^j \end{pmatrix},$$

so (U, φ^i, s_i) is also a coordinate system. In fact, it is a symplectic chart:

$$\begin{aligned} \omega|_U &= d\kappa + d\theta^i \wedge ds_i \\ &= dg^i \wedge ds_i + d\theta^i \wedge ds_i \\ &= d(g^i + \theta^i) \wedge ds_i \\ &= d\varphi^i \wedge ds_i. \end{aligned}$$

It only remains to find the integral curves of X_h in the action-angle coordinates (U, φ^i, s_i) . Observe that $\frac{\partial}{\partial \varphi^i} = X_{s_i}$ because for any $f \in \mathcal{C}^\infty(U)$ we have

$$X_{s_i} f = \{f, s_i\} = \frac{\partial f}{\partial \varphi^j} \frac{\partial s_i}{\partial s_j} + \frac{\partial f}{\partial s_j} \frac{\partial s_i}{\partial \varphi^j} = \frac{\partial f}{\partial \varphi^i}.$$

Therefore,

$$\frac{\partial h}{\partial \varphi^i} = X_{s_i} h = \{h, s_i\} = 0$$

(where we used that s_i is a constant of motion because it is a function of the integrals $f_1, \dots, f_n$). Hence, the Hamiltonian function h in action-angle variables is only a function of $s_1, \dots, s_n$. Moreover, the formula for the expression of a Hamiltonian field (Eq. (3.2)) gives

$$X_h = \frac{\partial h}{\partial s_i} \frac{\partial}{\partial \varphi^i}.$$

Defining $q^i = \frac{\partial h}{\partial s_i}$ (which depends only on $s_1, \dots, s_n$), it's clear that the integral curves are given by

$$\begin{cases} \dot{\varphi}^i = q^i(s_1, \dots, s_n) \\ \dot{s}_i = 0 \end{cases}$$

□

4.3 Consequences and examples

Now that the proof is complete, let us make some comments on action-angle variables. First, each compact regular level set has a neighborhood made up of invariant tori, so any trajectory of the Hamiltonian system starting in that neighborhood lies on a torus. Furthermore, the motion on the torus is given in angle coordinates by

$$\dot{\varphi}^i = q^i, \quad i = 1, \dots, n.$$

where the q^i , the so-called frequencies, are constants that depend only on the torus. The dynamics of this kind of differential equation on a torus is well-known (see, for instance, Example 4.1.11 of [1] and the section on conditionally periodic motion of [2]). Let us review this briefly.

In the case of the classical 2-dimensional torus, the solution of the system is

$$\begin{cases} \varphi^1(t) = q^1 t + \varphi^1(0) \\ \varphi^2(t) = q^2 t + \varphi^2(0) \end{cases} \quad (4.2)$$

(where both equations are modulo 2π). If the ratio q^1/q^2 is rational (including the case $q^2 = 0$), all trajectories are periodic. By contrast, if q^1/q^2 is irrational, none of the trajectories are periodic, and they are dense instead. This means that the topological closure of any trajectory is the whole torus, or equivalently that any trajectory crosses any neighborhood of any point on the torus. These trajectories are sometimes called *quasiperiodic orbits*.

The trajectories described in the system (4.2) can be visualized either in the plane, thinking of the torus as a square with opposite sides identified, or in the torus of revolution. Both approaches are applied to a periodic orbit in Figure 3. For an animated version of a periodic orbit and a dense orbit, visit https://github.com/rqdefrutos/masters-thesis/blob/main/torus/torus_anim_dense.gif and https://github.com/rqdefrutos/masters-thesis/blob/main/torus/torus_anim_periodic.gif.

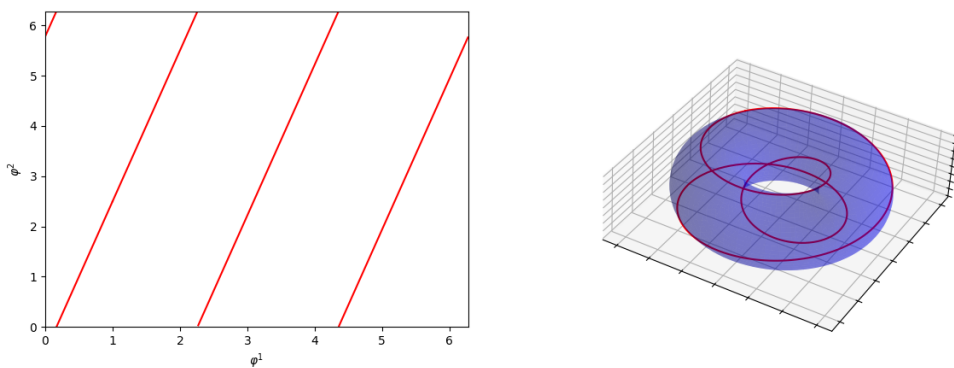


Figure 3: Rational winding of the torus with frequencies $q^1 = 1, q^2 = 3$. In the left, a periodic orbit is shown in the torus represented as a square of length 2π with opposite sides identified. Each axis represents an angle variable. In the right, the trajectory winds around the usual torus of revolution in $\mathbb{R}^3$.

Now, let us consider the general case of an n -torus. The so-called frequencies q^i are said to be

commensurable (or *rationaly dependent*) if there exist integers k_i , not all zero, such that

$$\sum_{i=1}^n k_i q^i = 0.$$

This is equivalent to the q^i being linearly dependent over the rational numbers, thus the name *rational dependence*. If the q^i are not commensurable, none of the trajectories are periodic, and they are dense on the torus. On the other hand, if the q^i are commensurable, each trajectory is contained in a submanifold of $\mathbb{T}^n$, which is actually a torus of lower dimension (see Figure 4). This submanifold could be all of $\mathbb{T}^n$. The trajectories can be either periodic or dense on the lower dimensional torus.

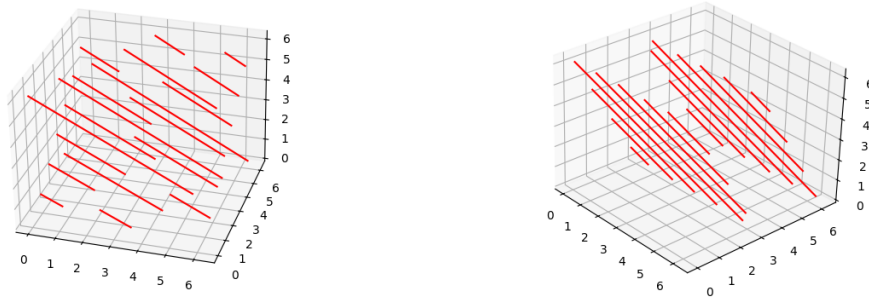


Figure 4: Two periodic orbits in the 3-torus $\mathbb{T}^3$ visualized as a cube where opposite sides are identified. As can be seen, the orbit on the right is contained in the quotient of a plane, which is a 2-torus.

Finally, let us illustrate the Liouville–Arnold theorem with a couple detailed examples.

Example 4.3.1. Consider an ideal pendulum fixed at the origin with mass m and radius R . The configuration space Q is the circle of radius R centered at the origin, so $Q \cong \mathbb{S}^1$, and the Hamiltonian phase space is

$$T^*Q \simeq T^*\mathbb{S}^1 \simeq \mathbb{S}^1 \times \mathbb{R},$$

i.e., a cylinder.

As is well-known, the Hamiltonian function of the pendulum with respect to the usual angle coordinate of the circle (measured from the negative part of the y -axis) is

$$h = \frac{p_\theta^2}{2mR^2} - mgR \cos \theta.$$

Hamilton's equations are

$$\begin{cases} \dot{\theta} = \frac{\partial h}{\partial p_\theta} \\ \dot{p}_\theta = -\frac{\partial h}{\partial \theta} \end{cases}$$

or more explicitly

$$\begin{cases} \dot{\theta} = \frac{p_\theta}{mR^2} \\ \dot{p}_\theta = -mgR \sin \theta \end{cases}$$

The trajectories of this system are shown in Figure 5. Note that

$$dh = mgR \sin \theta d\theta + \frac{p_\theta}{mR^2} dp_\theta,$$

which only vanishes in the set

$$\{(\theta, p_\theta) : \sin \theta = 0, p_\theta = 0\} = \{(\pi k, 0) : k \in \mathbb{Z}\}.$$

Since $dh \neq 0$ almost everywhere, the pendulum is an integrable system with h itself being the only integral.

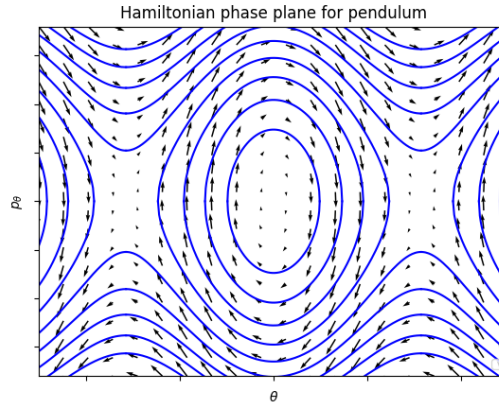


Figure 5: Hamiltonian phase plane of the ideal pendulum

Let's identify the level sets M_λ , which are given by the equation

$$\frac{p_\theta^2}{2mR^2} - mgR \cos \theta = \lambda.$$

Since $p_\theta^2 \geq 0$ and $\cos \theta \leq 1$,

$$\lambda \geq -mgR \cos \theta \geq -mgR.$$

Therefore, M_λ is the empty set for $\lambda < -mgR$. These are thus trivial regular values. Assume $\lambda \geq -mgR$ is a singular value. Then there exists $k \in \mathbb{Z}$ such that $(\pi k, 0) \in M_\lambda$, i.e.,

$$-mgR \cos(\pi k) = \lambda.$$

Since $\cos(\pi k) = (-1)^k$, the singular values of H are $\lambda = \pm mgR$.

Figure 6 (left) shows the Hamiltonian phase plane with M_λ for various values of λ . For $\lambda < -mgR$, it's empty, as we discussed before. For $\lambda = -mgR$ (which is a singular value), the level set consists of the equilibrium points $(\pi k, 0)$ with $k \in \mathbb{Z}$. For $-mgR < \lambda < mgR$, we get a sequence of circles. For $\lambda = mgR$, M_λ is not a smooth curve because λ is not regular. For $\lambda > mgR$, M_λ has two different connected components.

Projecting the trajectories onto the cylinder $\mathbb{S}^1 \times \mathbb{R}$, we see that the two connected components of M_λ for $\lambda > mgR$ are also circles, but these ones go around the cylinder, as shown in Figure 6 (right). In conclusion, in this case all the components of M_λ for regular values of λ are compact, and in accordance with Liouville–Arnold theorem, they are one-dimensional tori, i.e., circles.

Example 4.3.2. Consider the n -dimensional harmonic oscillator. We have seen in Example 4.1.4 that the Hamiltonian function is of the form

$$h = \frac{1}{2} \sum_{i=1}^n (a(q^i)^2 + bp_i^2)$$

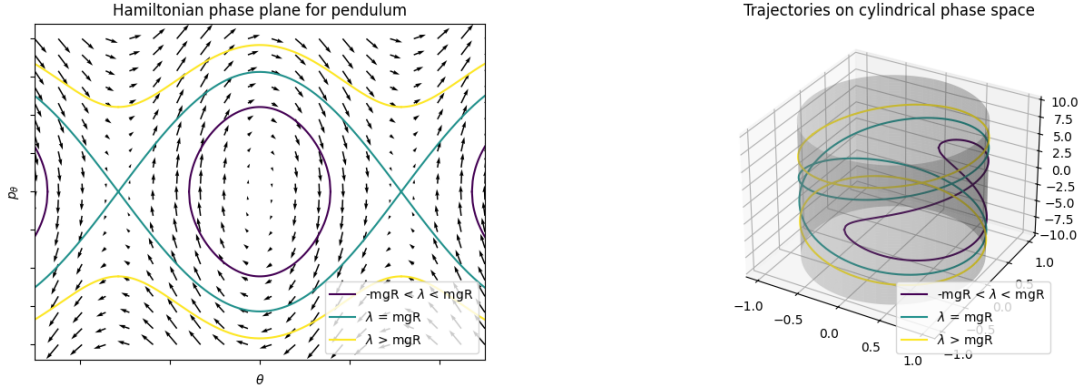


Figure 6: Level sets M_λ for some representative values of λ (left) and on the cylindrical phase space of the pendulum (right).

for some constants $a, b > 0$. For simplicity, we will take $a = b = 1$. We have

$$\frac{\partial h}{\partial q^i} = q^i, \quad \frac{\partial h}{\partial p_i} = p_i,$$

so the equations of motion are:

$$\begin{cases} \dot{q}^i = p_i \\ \dot{p}_i = -q^i \end{cases}$$

First of all, the level sets of the integrals $f_i = \frac{1}{2}(a(q^i)^2 + bp_i^2)$ are of the form

$$M_\Lambda = \{(q^1, \dots, q^n, p_1, \dots, p_n) \in \mathbb{R}^{2n} : (q^i)^2 + p_i^2 = 2\Lambda_i\}.$$

If all the Λ_i are positive, each equation describes a circle, so $M_\Lambda \simeq \mathbb{T}^n$.

Furthermore, a simple computation shows that (φ^i, f_i) , where

$$\varphi^i = \arctan\left(\frac{q^i}{p_i}\right),$$

is a set of action-angle coordinates. In fact, it is easy to check that they are Darboux coordinates, that is,

$$dq^i \wedge dp_i = d\varphi^i \wedge df_i,$$

and that the new equations of motion are:

$$\begin{cases} \dot{\varphi}^i = 1 \\ \dot{f}_i = 0 \end{cases}$$

5. Conclusions and outlook

We started this work reviewing how the language of differential geometry allows one to express the Lagrangian and Hamiltonian formalisms of mechanics in a conceptual and modern way. This set the ground for Chapter 3, where we covered the fundamentals of symplectic geometry and showed why it is a natural language for Hamiltonian mechanics. This involved reviewing key concepts of symplectic manifolds, such as the Darboux theorem, Hamiltonian vector fields and Poisson brackets.

After establishing the framework of symplectic mechanics, we introduced the Liouville–Arnold theorem, which is the central result of the thesis. We provided a highly detailed proof of the theorem, which required using several techniques, such as group actions, locally trivial fibrations, topological arguments, and manipulation of differential forms under successive changes of variables. In addition, we illustrated the result with various examples, some of which are accompanied by computer-generated graphics that help visualize the geometric aspect of the theorem.

Finally, let us make some comments on possible future work. The natural step after studying integrability in symplectic geometry would be its generalization to other geometric structures, such as cosymplectic or contact geometry. Integrable systems and the Liouville–Arnold theorem can also be stated more generally, as shown, for example, in [5, 16]. There are also more algebraic approaches to integrability involving Nijenhuis and Haantjes endomorphisms, which would also be interesting to explore. Lastly, another direction for further work is the Kolmogorov–Arnold–Moser theory concerning the persistence of quasiperiodic motion in integrable Hamiltonian systems under perturbations.

A. Ehresmann fibration theorem

We devote this appendix to prove a result that is used in the proof of the Liouville–Arnold theorem. Essentially, it is a local version of Ehresmann fibration theorem. Before, we need some preliminaries.

Recall that, given a smooth map $F : M \rightarrow N$ and two vector fields $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$, we say that X and Y are F -related if

$$T_p F(X(p)) = Y(F(p)), \quad p \in M.$$

For example, if F is a diffeomorphism, then X and Y are F -related if and only if Y is the pushforward of X , denoted $Y = F_* X$. For the case in which F is a submersion, we have the following lemma about the existence of F -related fields.

Lemma A.1. *Let $F : M \rightarrow N$ be a submersion. For every field $Y \in \mathfrak{X}(N)$ there exists a field $X \in \mathfrak{X}(M)$ such that X and Y are F -related.*

Proof. Let m and n be the dimensions of M and N , respectively. Note that $m \geq n$. For any $p \in M$ there are charts (U, x^i) around p and (V, y^i) around $F(p)$ such that $F(U) \subset V$ and the local expression of F is

$$y \circ F \circ x^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^n).$$

Observing that for any $p \in U$ the matrix of $T_p F$ with respect to these coordinates is $(I_n \mid 0_{n \times (m-n)})$, it is immediate that

$$T_p F \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial}{\partial y^i} \Big|_{F(p)}, \quad i = 1, \dots, n.$$

In other words, the partial derivatives $\frac{\partial}{\partial x^i}$ and $\frac{\partial}{\partial y^i}$ are F -related. Thus, for any field $Y \in \mathfrak{X}(N)$ with local expression $Y|_V = Y^i \frac{\partial}{\partial y^i}$, the field $X = Y^i \circ F \frac{\partial}{\partial x^i} \in \mathfrak{X}(U)$ satisfies

$$T_p F(X(p)) = T_p F \left(Y^i(F(p)) \frac{\partial}{\partial x^i} \Big|_p \right) = Y^i(F(p)) \frac{\partial}{\partial y^i} \Big|_{F(p)} = Y(F(p)).$$

This local construction can be made global as follows. We cover M with charts $\{U_\alpha\}$ satisfying that, for any $Y \in \mathfrak{X}(N)$, there are fields $X_\alpha \in \mathfrak{X}(U_\alpha)$ such that $T_p F(X_\alpha(p)) = Y(F(p))$ for $p \in U_\alpha$. Now, consider a partition of unity $\{f_\alpha\}$ subordinate to this cover. We claim that the global field $X = \sum_\alpha f_\alpha X_\alpha$ is F -related to Y . Indeed, for any $p \in M$,

$$T_p F(X(p)) = T_p F \left(\sum_\alpha f_\alpha(p) X_\alpha(p) \right) = \sum_\alpha f_\alpha(p) Y(F(p)) = Y(F(p))$$

because $\sum_\alpha f_\alpha(p) = 1$.

□

An elementary property of F -related fields is that their flows, ϕ_X^t and ϕ_Y^t , satisfy $F \circ \phi_X^t = \phi_Y^t \circ F$ whenever both sides are defined. To check this, fix p and consider the curve $c(t) = F(\phi_X^t(p))$. We have

$$\frac{d}{dt} F(\phi_X^t(p)) = T_{\phi_X^t(p)} F \left(\frac{d}{dt} \phi_X^t(p) \right) = T_{\phi_X^t(p)} F(X(\phi_X^t(p))) = Y(F(\phi_X^t(p))),$$

so $c(t)$ is an integral curve of Y . Since $c(0) = F(p)$, it must be $c(t) = \phi_Y^t(F(p))$. In other words, $F(\phi_X^t(p)) = \phi_Y^t(F(p))$.

Before proving the theorem, we need an easy lemma about compactness.

Lemma A.2. *Let $X \in \mathfrak{X}(M)$ be a vector field on a manifold M and let $K \subset M$ be a compact set. There exists $\varepsilon > 0$ such that the flow $\phi^t(p)$ of X is defined for $p \in K$, $|t| < \varepsilon$.*

Proof. The flow of X is a map $\phi : W \rightarrow M$ where W is an open set of $\mathbb{R} \times M$ such that $\{0\} \times M \subset W$. The statement is then equivalent to $(-\varepsilon, \varepsilon) \times K \subset W$.

Since $(0, p) \in W$ for every $p \in K$ and W is open, there exists $\varepsilon_p > 0$ and an open neighborhood U_p of p such that $(-\varepsilon_p, \varepsilon_p) \times U_p \subset W$. The U_p form an open cover of K . Since K is compact, there are $p_1, \dots, p_k \in K$ such that

$$K \subset U_{p_1} \cup \dots \cup U_{p_k}.$$

Let $\varepsilon = \min\{\varepsilon_{p_1}, \dots, \varepsilon_{p_k}\}$ and let's check that $(-\varepsilon, \varepsilon) \times K \subset W$. If $(t, p) \in (-\varepsilon, \varepsilon) \times K$, there is some index i such that $p \in U_{p_i}$, so

$$(t, p) \in (-\varepsilon_{p_i}, \varepsilon_{p_i}) \times U_{p_i} \subset W.$$

□

We are now ready to prove the following theorem.

Theorem A.3. *Let $F : M \rightarrow N$ be a smooth map and $y_0 \in N$ a regular value such that $F^{-1}(y_0)$ is compact. Then there exists an open neighborhood V of y_0 and a diffeomorphism $\Phi : F^{-1}(y_0) \times V \rightarrow F^{-1}(V)$ such that $F \circ \Phi : F^{-1}(y_0) \times V \rightarrow V$ is the projection onto the second factor.*

Proof. Without loss of generality, we can assume $N = \mathbb{R}^n$ and $y_0 = 0$. Since 0 is a regular value and the set of regular points of F is open, there exists an open neighborhood U of $F^{-1}(0)$ such that $T_p F$ is surjective for every $p \in U$, i.e. the restriction $F|_U : U \rightarrow \mathbb{R}^n$ is a submersion.

Consider the coordinate fields $\frac{\partial}{\partial y^i} \in \mathfrak{X}(\mathbb{R}^n)$. By Lemma A.1, there exist fields $X_i \in \mathfrak{X}(U)$ such that $\frac{\partial}{\partial y^i}$ and X_i are F -related. The flow of X_i is a map $\phi_i : W_i \subset \mathbb{R} \times U \rightarrow U$ where W_i is an open set with $\{0\} \times U \subset W_i$. By Lemma A.2, we have $(-\varepsilon_i, \varepsilon_i) \times F^{-1}(0) \subset W_i$ for some $\varepsilon_i > 0$. Let $V = (-\varepsilon_1, \varepsilon_1) \times \dots \times (-\varepsilon_n, \varepsilon_n) \subset \mathbb{R}^n$, which is an open neighborhood of 0. By construction, the smooth map

$$\Phi : F^{-1}(0) \times V \rightarrow U, (p, \mathbf{y}) \mapsto \phi_1^{y^1} \circ \dots \circ \phi_n^{y^n}(p)$$

is well-defined.

On the other hand, the flow of $\frac{\partial}{\partial y^i}$ is clearly

$$\psi_i^t(\mathbf{x}) = \mathbf{x} + t\mathbf{e}_i,$$

where $\mathbf{e}_i$ is the i -th vector of the canonical basis of $\mathbb{R}^n$. Since X_i and $\frac{\partial}{\partial y^i}$ are F -related, we have $F \circ \phi_i^t = \psi_i^t \circ F$. Using this, let us show that $F \circ \Phi(p, \mathbf{y}) = \mathbf{y}$, i.e., $F \circ \Phi$ is the projection onto the second factor:

$$\begin{aligned} F \circ \Phi(p, \mathbf{y}) &= F \circ \phi_1^{y^1} \circ \dots \circ \phi_n^{y^n}(p) = \psi_1^{y^1} \circ \dots \circ \psi_n^{y^n} \circ F(p) \\ &= \psi_1^{y^1} \circ \dots \circ \psi_n^{y^n}(0) = \psi_1^{y^1} \circ \dots \circ \psi_{n-1}^{y^{n-1}}(y^n \mathbf{e}_n) \\ &= \psi_1^{y^1} \circ \dots \circ \psi_{n-2}^{y^{n-2}}(y^{n-1} \mathbf{e}_{n-1} + y^n \mathbf{e}_n) = \dots = \mathbf{y} \end{aligned}$$

In particular, this shows that the image of Φ is contained in $F^{-1}(V)$, so Φ is actually a map $\Phi : F^{-1}(0) \times V \rightarrow F^{-1}(V)$. Finally, it is immediate that Φ is a diffeomorphism because its inverse is

$$\Phi^{-1}(q) = (\phi_n^{-y^n} \circ \dots \circ \phi_1^{-y^1}(q), F(q))$$

where $(y^1, \dots, y^n) = F(q)$. □

Remark A.4. Restricting Φ gives a diffeomorphism $F^{-1}(y_0) \times \{y\} \simeq F^{-1}(y)$, so all fibers $F^{-1}(y)$ with $y \in V$ are diffeomorphic to $F^{-1}(y_0)$.

Remark A.5. The conclusion of the theorem is summarized by saying that F is *locally trivial at* y_0 . An interesting way to rephrase the condition of being locally trivial is that there exists a map $\eta : F^{-1}(V) \rightarrow F^{-1}(y_0)$ such that $(\eta, F) : F^{-1}(V) \rightarrow F^{-1}(y_0) \times V$ is a diffeomorphism.

Remark A.6. We can always replace V with a smaller neighborhood of y_0 . For instance, we can take V to be diffeomorphic to an open disk.

When F is locally trivial at every point of its image, it is said that F is a *locally trivial fibration*. In particular, if F is a proper map (meaning that the preimage of a compact set is compact, so every fiber $F^{-1}(y_0)$ is compact) and F is a submersion (so every y_0 is a regular value), then Theorem [A.3](#) ensures that F is a locally trivial fibration. This is the famous Ehresmann fibration theorem:

Corollary A.7. (*Ehresmann*) *Every proper submersion is a locally trivial fibration.*

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